

# Complete Calculation Record

One-Dimensional Guyer–Krumhansl Heat Conduction  
as an Inverse Problem for  $(\alpha, \tau_q, \kappa^2)$

Technical calculation document — not a manuscript

23 August 2026

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**Purpose and status of this document.** This document records every calculation underlying the present study in full, including intermediate algebra that would normally be compressed. It is written so that the program of Phase 3 can be implemented directly from it, and so that the program output of Phase 4 can be compared against it line by line in Phase 5. It contains no introduction, no discussion, no conclusions and no novelty claims.

**Provenance labelling.** Every quantitative statement carries one of four labels, and these are never mixed:

- `[LITERATURE VALUE]` a value taken from a published source, cited at the point of use;
- `[DERIVED VALUE]` a value obtained by algebra from literature values or from equations derived in this document;
- `[NUMERICALLY COMPUTED HERE]` a value computed numerically in this work, with the script named;
- `[PREVIOUSLY VERIFIED]` a value established in earlier verified work and not recomputed in this pass; explicitly flagged as such.

**Reproducibility.** Every numerical result quoted below names the script that produced it. Scripts live in `validation/calc/` and `repro/`. The audit records are `checks_symbolic.json` (30 symbolic and dimensional checks) and `checks_numerical.json` (19 numerical checks), both in `validation/calc/`.

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## Contents

|          |                                                               |          |
|----------|---------------------------------------------------------------|----------|
| <b>1</b> | <b>Problem definition</b>                                     | <b>4</b> |
| 1.1      | Explicit exclusion list . . . . .                             | 4        |
| <b>2</b> | <b>Dimensional Guyer–Krumhansl equations</b>                  | <b>5</b> |
| 2.1      | Energy balance . . . . .                                      | 5        |
| 2.2      | Constitutive law . . . . .                                    | 5        |
| 2.3      | Notation warning: $\kappa^2$ versus a length scale . . . . .  | 6        |
| 2.4      | Elimination of $T$ : single equation for the flux . . . . .   | 6        |
| 2.5      | Symbol table . . . . .                                        | 7        |
| <b>3</b> | <b>Initial and boundary conditions</b>                        | <b>7</b> |
| 3.1      | The initial time-derivative is derived, not imposed . . . . . | 8        |
| 3.2      | Choice of representation . . . . .                            | 8        |
| <b>4</b> | <b>Nondimensionalisation</b>                                  | <b>8</b> |
| 4.1      | Characteristic scales and substitutions . . . . .             | 8        |
| 4.2      | Line-by-line transformation of the field equations . . . . .  | 9        |

|           |                                                              |           |
|-----------|--------------------------------------------------------------|-----------|
| <b>5</b>  | <b>Laplace-domain formulation and the propagation factor</b> | <b>10</b> |
| <b>6</b>  | <b>Definition and interpretation of <math>B</math></b>       | <b>12</b> |
| 6.1       | Reparameterisation . . . . .                                 | 13        |
| <b>7</b>  | <b>Limiting cases</b>                                        | <b>14</b> |
| <b>8</b>  | <b>The Fourier-resonance condition</b>                       | <b>15</b> |
| <b>9</b>  | <b>Exact structural identifiability</b>                      | <b>15</b> |
| 9.1       | Complete dependence of the forward solution . . . . .        | 15        |
| 9.2       | Sensitivities . . . . .                                      | 16        |
| 9.3       | Jacobian rank, Fisher singularity, correlation . . . . .     | 17        |
| <b>10</b> | <b>Numerical proof that the degeneracy is exact</b>          | <b>18</b> |
| <b>11</b> | <b>Sensitivity matrix and Fisher information</b>             | <b>19</b> |
| <b>12</b> | <b>Reparameterisation: what remains determined</b>           | <b>20</b> |
| 12.1      | Numerical evidence, kept separate from the theorem . . . . . | 21        |
| <b>13</b> | <b>Practical identifiability</b>                             | <b>21</b> |
| 13.1      | Derivation of the band boundaries . . . . .                  | 22        |
| <b>14</b> | <b>Profile likelihood</b>                                    | <b>23</b> |
| <b>15</b> | <b>Monte Carlo parameter recovery</b>                        | <b>24</b> |
| <b>16</b> | <b>Model selection by BIC</b>                                | <b>25</b> |
| <b>17</b> | <b>Published calibration calculations</b>                    | <b>26</b> |
| 17.1      | Worked arithmetic for each case . . . . .                    | 26        |
| 17.2      | The Ván et al. $\ell/b$ consistency question . . . . .       | 27        |
| 17.3      | Master calibration table . . . . .                           | 28        |
| 17.4      | Same-specimen discrepancy . . . . .                          | 28        |
| <b>18</b> | <b>The Fehér–Kovács fitting domain</b>                       | <b>29</b> |
| <b>19</b> | <b>Calibration uncertainty consistency check</b>             | <b>29</b> |
| <b>20</b> | <b>Validation anchor: reproduction of published results</b>  | <b>30</b> |
| 20.1      | Truncated eigenfunction series required by Fig. 2 . . . . .  | 30        |
| <b>21</b> | <b>Numerical certification</b>                               | <b>32</b> |
| 21.1      | Factorisation and why the pulse is not split . . . . .       | 32        |
| 21.2      | Convergence, precision and cross-solver checks . . . . .     | 33        |
| <b>22</b> | <b>Master traceability table</b>                             | <b>34</b> |
| <b>23</b> | <b>Program implementation specification</b>                  | <b>35</b> |
| 23.1      | Module 1: forward solver . . . . .                           | 35        |
| 23.2      | Module 2: analytic Jacobian . . . . .                        | 36        |
| 23.3      | Module 3: independent cross-check solver . . . . .           | 36        |
| 23.4      | Module 4: estimation and statistics . . . . .                | 36        |
| 23.5      | Required validation tests and expected values . . . . .      | 37        |
| 23.6      | Required figures . . . . .                                   | 37        |

|                                                                   |           |
|-------------------------------------------------------------------|-----------|
| 23.7 Output format . . . . .                                      | 37        |
| <b>24 Final calculation audit</b>                                 | <b>38</b> |
| 24.1 Discrepancies found and resolved during this audit . . . . . | 38        |
| 24.2 Stale-number scan . . . . .                                  | 38        |
| 24.3 Remaining blockers . . . . .                                 | 38        |

# 1 Problem definition

**Definition 1.1** (Physical problem). A homogeneous, rigid, isotropic slab of thickness  $L$  is initially in thermal equilibrium with its surroundings. At  $t = 0$  the face  $x = 0$  receives a heat pulse of finite duration  $t_p$ . The opposite face  $x = L$  is thermally insulated. The measured quantity is the temperature history of the rear face,  $T(L, t)$ . This is the laser flash configuration.

**Definition 1.2** (Domain and fields).

$$\text{spatial domain: } x \in [0, L], \quad L > 0, \quad (1)$$

$$\text{time domain: } t \geq 0, \quad (2)$$

$$\text{temperature field: } T(x, t) \quad [\text{K}], \quad (3)$$

$$\text{heat flux field: } q(x, t) \quad [\text{W m}^{-2}]. \quad (4)$$

The problem is strictly one-dimensional: no quantity depends on  $y$  or  $z$ , and  $q$  denotes the  $x$ -component of the flux vector.

**Definition 1.3** (Material parameters). The medium is characterised by four constants, all positive:

$$\rho c \quad \text{volumetric heat capacity} \quad [\text{J m}^{-3} \text{K}^{-1}], \quad (5)$$

$$\lambda \quad \text{thermal conductivity} \quad [\text{W m}^{-1} \text{K}^{-1}], \quad (6)$$

$$\tau_q \quad \text{relaxation time of the heat flux} \quad [\text{s}], \quad (7)$$

$$\kappa^2 \quad \text{Guyer–Krumhansl nonlocal coefficient} \quad [\text{m}^2]. \quad (8)$$

The thermal diffusivity is the derived combination  $\alpha = \lambda/(\rho c)$ , in  $\text{m}^2 \text{s}^{-1}$ .

**Assumption 1.1** (Constitutive assumptions). (A1) The solid is rigid: no mechanical deformation, no coupling of the temperature field to a displacement field.

(A2)  $\rho$ ,  $c$ ,  $\lambda$ ,  $\tau_q$  and  $\kappa^2$  are constants, independent of temperature and position.

(A3) There is no internal heat generation.

(A4) The heat flux obeys the Guyer–Krumhansl constitutive law (11) below, which is linear in  $q$ ,  $\partial_t q$ ,  $\partial_{xx} q$  and  $\partial_x T$ .

(A5) All coefficients are non-negative, as required for thermodynamic admissibility.

**Definition 1.4** (Inverse problem). The forward problem maps  $(\alpha, \tau_q, \kappa^2)$  to the rear-face history  $T(L, t)$ . The inverse problem is the recovery of the parameter vector

$$\boldsymbol{\theta} = (\alpha, \tau_q, \kappa^2) \quad (9)$$

from a finite, noisy sample of  $T(L, t)$ . The central question of the calculations below is which components or combinations of  $\boldsymbol{\theta}$  are determined by that observation.

## 1.1 Explicit exclusion list

The following are *not* part of the model and appear nowhere in this document or in the program to be built from it. This list is fixed and is not to be extended later.

| Excluded                       | Reason                                                                          |
|--------------------------------|---------------------------------------------------------------------------------|
| Thermoelastic coupling         | rigid conductor, Assumption (A1)                                                |
| Piezoelectric coupling         | no electric field in the model                                                  |
| Two-temperature theory         | single temperature field, (3)                                                   |
| Fractional-order derivatives   | constitutive law is integer order                                               |
| Memory-dependent derivatives   | no memory kernel                                                                |
| Rabotnov kernels               | no memory kernel                                                                |
| Porosity as a mechanism        | effective homogeneous medium                                                    |
| Semiconductor carrier coupling | no carrier transport equation                                                   |
| Klein–Gordon nonlocal terms    | not present in (11)                                                             |
| Nonlocal elasticity (Eringen)  | no mechanical field                                                             |
| Moore–Gibson–Thompson (MGT)    | requires a thermal-displacement variable absent from the transfer function (44) |
| Multi-dimensional geometry     | strictly 1-D, definition 1.2                                                    |

## 2 Dimensional Guyer–Krumhansl equations

### 2.1 Energy balance

**Derivation 2.1** (Balance of internal energy). For a rigid conductor the specific internal energy is  $e = cT$ . Conservation of energy in one dimension, with no source term, reads  $\rho \partial_t e + \partial_x q = 0$ . Substituting  $e = cT$  with  $c$  constant gives

$$\boxed{\rho c \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0} \quad (10)$$

**SOURCE.** Standard balance law for a rigid heat conductor; used in exactly this form by the validation anchor [1] and by [2, 3]. Not specific to any constitutive assumption.

**CHECK.** Term-by-term dimensions of (10):

$$\begin{aligned} [\rho c \partial_t T] &= \text{J m}^{-3} \text{K}^{-1} \cdot \text{K s}^{-1} = \text{J m}^{-3} \text{s}^{-1} = \text{W m}^{-3}, \\ [\partial_x q] &= \text{W m}^{-2} \cdot \text{m}^{-1} = \text{W m}^{-3}. \end{aligned}$$

Both terms are  $\text{W m}^{-3}$ . Dimensionally consistent.

### 2.2 Constitutive law

**Derivation 2.2** (Guyer–Krumhansl constitutive equation). The Guyer–Krumhansl law extends the Maxwell–Cattaneo–Vernotte relation  $\tau_q \partial_t q + q + \lambda \partial_x T = 0$  by a term proportional to the second spatial derivative of the flux:

$$\boxed{\tau_q \frac{\partial q}{\partial t} + q + \lambda \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0} \quad (11)$$

**SOURCE.** Originally obtained by Guyer and Krumhansl from the linearised phonon Boltzmann equation for dielectric crystals [4, 5]. An equivalent structure follows from non-equilibrium thermodynamics with internal variables and entropy-current multipliers, in which the coefficients are constrained only by the second law and no carrier mechanism is assumed [6, 7]. The flux form (11) is the form used by the validation anchor [1].

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**CHECK.** Term-by-term dimensions of (11), target  $\text{W m}^{-2}$ :

$$\begin{aligned} [\tau_q \partial_t q] &= \text{s} \cdot \text{W m}^{-2} \text{s}^{-1} = \text{W m}^{-2}, \\ [q] &= \text{W m}^{-2}, \\ [\lambda \partial_x T] &= \text{W m}^{-1} \text{K}^{-1} \cdot \text{K m}^{-1} = \text{W m}^{-2}, \\ [\kappa^2 \partial_{xx} q] &= \text{m}^2 \cdot \text{W m}^{-2} \cdot \text{m}^{-2} = \text{W m}^{-2}. \end{aligned}$$

All four terms are  $\text{W m}^{-2}$ . Dimensionally consistent, and  $\kappa^2$  must have dimension  $\text{m}^2$  for this to hold; this fixes the meaning of the symbol.

---

### 2.3 Notation warning: $\kappa^2$ versus a length scale

Different sources parameterise the nonlocal term differently, and the conversions must be stated rather than assumed.

- This document and the anchor [1] use  $\kappa^2$  with dimension  $\text{m}^2$ , multiplying  $\partial_{xx} q$  in (11).
- Ván et al. [3] write the same term with  $\ell^2$ , where  $\ell$  is a length, so that  $\ell^2 \equiv \kappa^2$  dimensionally, and additionally report the dimensionless group  $b = \ell^2/(\tau\alpha)$ .
- Fehér and Kovács [8] tabulate  $\kappa^2$  directly in  $\text{m}^2$  and quote the ratio  $\kappa^2/(\alpha\tau_q)$ .

---

**RESULT.**  $b$  of [3] and  $\kappa^2/(\alpha\tau_q)$  of [8] denote the same dimensionless quantity, called  $B$  in section 6. No symbol from one source is identified with a similarly named symbol from another without the explicit conversion above. Where a source tabulates  $\ell$  rather than  $\kappa^2$ , the value actually used in section 17 is the one the authors themselves report, and any inconsistency is flagged rather than silently recomputed.

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### 2.4 Elimination of $T$ : single equation for the flux

**Derivation 2.3** (Flux equation). Differentiate (11) with respect to  $x$ :

$$\tau_q \frac{\partial^2 q}{\partial t \partial x} + \frac{\partial q}{\partial x} + \lambda \frac{\partial^2 T}{\partial x^2} - \kappa^2 \frac{\partial^3 q}{\partial x^3} = 0. \quad (12)$$

From the energy balance (10),

$$\frac{\partial T}{\partial t} = -\frac{1}{\rho c} \frac{\partial q}{\partial x} \quad \implies \quad \frac{\partial^2 T}{\partial x^2} = -\frac{1}{\rho c} \frac{\partial}{\partial t} \left( \frac{\partial q}{\partial x} \right)^* \quad (13)$$

where the starred step uses the interchange of  $\partial_{xx}$  and  $\partial_t$  applied to (10) integrated once in  $x$ ; more directly, differentiating (10) with respect to  $x$  gives  $\rho c \partial_t \partial_x T = -\partial_{xx} q$ , so  $\partial_x T$  satisfies  $\partial_t(\partial_x T) = -\partial_{xx} q/(\rho c)$ . Substituting  $\lambda \partial_{xx} T$  from this relation into (12) is most cleanly done in the Laplace domain, which is why section 5 performs the elimination there. Carrying it out in the time domain for zero initial data gives the equivalent single equation

$$\boxed{\tau_q \frac{\partial^2 q}{\partial t^2} + \frac{\partial q}{\partial t} = \alpha \frac{\partial^2 q}{\partial x^2} + \kappa^2 \frac{\partial^3 q}{\partial t \partial x^2}} \quad (14)$$

with  $\alpha = \lambda/(\rho c)$ .

---

**CHECK.** Dimensions of (14), target  $\text{W m}^{-2} \text{s}^{-1}$ :  $[\tau_q \partial_{tt} q] = \text{s} \cdot \text{W m}^{-2} \text{s}^{-2} = \text{W m}^{-2} \text{s}^{-1}$ ;  $[\partial_t q] = \text{W m}^{-2} \text{s}^{-1}$ ;  $[\alpha \partial_{xx} q] = \text{m}^2 \text{s}^{-1} \cdot \text{W m}^{-2} \text{m}^{-2} = \text{W m}^{-2} \text{s}^{-1}$ ;  $[\kappa^2 \partial_t \partial_{xx} q] = \text{m}^2 \cdot \text{W m}^{-2} \text{s}^{-1} \text{m}^{-2} = \text{W m}^{-2} \text{s}^{-1}$ . All four terms agree. Consistent.

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**RESULT.** The same elimination performed on  $T$  instead of  $q$  produces an equation of identical form in  $T$ , namely  $\tau_q \partial_{tt} T + \partial_t T = \alpha \partial_{xx} T + \kappa^2 \partial_t \partial_{xx} T$  [1]. The two representations are *not* interchangeable once boundary data are imposed: see section 3.2.

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## 2.5 Symbol table

Table 1: Symbols, meanings, SI units and source.

| Symbol           | SI unit                          | Meaning and source                                           |
|------------------|----------------------------------|--------------------------------------------------------------|
| $T$              | K                                | temperature field, definition 1.2                            |
| $q$              | $\text{W m}^{-2}$                | heat flux ( $x$ -component)                                  |
| $x$              | m                                | position, $0 \leq x \leq L$                                  |
| $t$              | s                                | time                                                         |
| $L$              | m                                | slab thickness                                               |
| $t_p$            | s                                | pulse duration; $t_p = 0.01$ s in the anchor experiments [1] |
| $\rho$           | $\text{kg m}^{-3}$               | mass density                                                 |
| $c$              | $\text{J kg}^{-1} \text{K}^{-1}$ | specific heat                                                |
| $\rho c$         | $\text{J m}^{-3} \text{K}^{-1}$  | volumetric heat capacity                                     |
| $\lambda$        | $\text{W m}^{-1} \text{K}^{-1}$  | thermal conductivity                                         |
| $\alpha$         | $\text{m}^2 \text{s}^{-1}$       | thermal diffusivity, $\alpha = \lambda/(\rho c)$             |
| $\tau_q$         | s                                | flux relaxation time, (11)                                   |
| $\kappa^2$       | $\text{m}^2$                     | nonlocal coefficient, (11)                                   |
| $B$              | —                                | resonance number $\kappa^2/(\alpha \tau_q)$ , section 6      |
| $q_{\max}$       | $\text{W m}^{-2}$                | pulse amplitude                                              |
| $\bar{q}_0$      | $\text{W m}^{-2}$                | time-averaged pulse flux                                     |
| $T_0$            | K                                | initial (ambient) temperature                                |
| $T_{\text{end}}$ | K                                | adiabatic final temperature                                  |
| $s$              | $\text{s}^{-1}$                  | Laplace variable conjugate to $t$                            |
| $m(s)$           | $\text{m}^{-1}$                  | propagation factor, (44)                                     |
| $\sigma$         | —                                | noise standard deviation (dimensionless $T$ )                |

---

## 3 Initial and boundary conditions

**Definition 3.1** (Excitation). The irradiated face receives a pulse of finite duration  $t_p$  and cosine shape,

$$q(0, t) = \begin{cases} q_{\max} \left[ 1 - \cos\left(\frac{2\pi t}{t_p}\right) \right], & 0 < t \leq t_p, \\ 0, & t > t_p. \end{cases} \quad (15)$$

**SOURCE.** This is exactly the boundary function of the anchor [1], which states  $t_p = 0.01$  s for the underlying experiments [2, 3]. It is not invented here and is not modified.

**Definition 3.2** (Rear face and initial state).

$$q(L, t) = 0 \quad \text{for all } t \quad (\text{adiabatic rear face}), \quad (16)$$

$$T(x, 0) = T_0, \quad (17)$$

$$q(x, 0) = 0. \quad (18)$$

Boundary cooling is neglected, as in the anchor [1].

### 3.1 The initial time-derivative is derived, not imposed

**Derivation 3.1** (Consistency of the second initial condition). Equation (14) is second order in time, so a well-posed problem needs both  $q(x, 0)$  and  $\partial_t q(x, 0)$ . These cannot be prescribed independently. Evaluating the constitutive law (11) at  $t = 0$  and solving for the time derivative,

$$\tau_q \frac{\partial q}{\partial t}(x, 0) = -q(x, 0) - \lambda \frac{\partial T}{\partial x}(x, 0) + \kappa^2 \frac{\partial^2 q}{\partial x^2}(x, 0). \quad (19)$$

With the quiescent state (17)–(18):  $q(x, 0) = 0$  gives  $\partial_{xx} q(x, 0) = 0$ , and  $T(x, 0) = T_0$  constant gives  $\partial_x T(x, 0) = 0$ . Hence every term on the right vanishes and

$$\frac{\partial q}{\partial t}(x, 0) = 0. \quad (20)$$

---

**RESULT.**  $\partial_t q(x, 0) = 0$  is a *consequence* of (17), (18) and the constitutive law, not an independent assumption. Imposing any other value would violate (11) at  $t = 0$ . The program of section 23 must not treat it as a free input.

---

### 3.2 Choice of representation

**Assumption 3.1** (Flux representation). The calculations use the flux representation (14), recovering  $T$  afterwards by integrating (10) in time.

**Reason.** The experiment prescribes the *flux* at  $x = 0$ , equation (15). In the temperature representation this boundary datum is not expressible in  $T$  alone without introducing time derivatives of the boundary temperature. The temperature representation is convenient only for Dirichlet data in  $T$ . The anchor makes the same choice for the same reason [1, 9].

## 4 Nondimensionalisation

### 4.1 Characteristic scales and substitutions

**Definition 4.1** (Scaling).

$$\hat{t} = \frac{\alpha t}{L^2}, \quad \hat{x} = \frac{x}{L}, \quad \hat{T} = \frac{T - T_0}{T_{\text{end}} - T_0}, \quad \hat{q} = \frac{q}{\bar{q}_0}. \quad (21)$$

**SOURCE.** Exactly the scaling of the anchor [1]. Reproduced here in full so every coefficient below can be checked independently.

**Calculation 4.1** (The averaged flux and the end temperature). The time-averaged pulse flux is

$$\begin{aligned} \bar{q}_0 &= \frac{1}{t_p} \int_0^{t_p} q(0, t) dt = \frac{q_{\text{max}}}{t_p} \int_0^{t_p} \left[ 1 - \cos\left(\frac{2\pi t}{t_p}\right) \right] dt \\ &= \frac{q_{\text{max}}}{t_p} \left[ t - \frac{t_p}{2\pi} \sin\left(\frac{2\pi t}{t_p}\right) \right]_0^{t_p} = \frac{q_{\text{max}}}{t_p} \left[ t_p - \frac{t_p}{2\pi} \sin(2\pi) \right] = q_{\text{max}}, \end{aligned} \quad (22)$$

since  $\sin 2\pi = 0$ . The energy deposited per unit area is  $\bar{q}_0 t_p$ ; spread uniformly over the insulated slab of heat capacity  $\rho c L$  per unit area it raises the temperature by  $\bar{q}_0 t_p / (\rho c L)$ , so

$$T_{\text{end}} = T_0 + \frac{\bar{q}_0 t_p}{\rho c L}. \quad (23)$$



---

**CHECK.**  $[\bar{q}_0 t_p / (\rho c L)] = \text{W m}^{-2} \cdot \text{s} \cdot \text{m}^3 \text{K J}^{-1} \cdot \text{m}^{-1} = \text{J m}^{-2} \cdot \text{m}^2 \text{K J}^{-1} = \text{K}$ . Correct. Consequently  $\hat{T} \rightarrow 1$  as  $\hat{t} \rightarrow \infty$ , which is the normalisation used for all reported temperature histories. Symbolic verification of (22): check 7d of `check_symbolic.py`, result  $\bar{q}_0/q_{\max} = 1$  exactly.

---

**Definition 4.2** (Dimensionless parameters).

$$\hat{\tau}_\Delta = \frac{\alpha t_p}{L^2}, \quad \hat{\tau}_q = \frac{\alpha \tau_q}{L^2}, \quad \hat{\kappa}^2 = \frac{\kappa^2}{L^2}. \quad (24)$$

---

**CHECK.** Each is dimensionless:  $[\alpha t_p / L^2] = \text{m}^2 \text{s}^{-1} \cdot \text{s} \cdot \text{m}^{-2} = 1$ ;  $[\alpha \tau_q / L^2] = 1$  identically;  $[\kappa^2 / L^2] = \text{m}^2 / \text{m}^2 = 1$ .

---

## 4.2 Line-by-line transformation of the field equations

**Derivation 4.1** (Energy balance in dimensionless variables). Write  $T = T_0 + (T_{\text{end}} - T_0)\hat{T}$  and  $q = \bar{q}_0 \hat{q}$ , with  $t = L^2 \hat{t} / \alpha$  and  $x = L \hat{x}$ . Then

$$\frac{\partial T}{\partial t} = (T_{\text{end}} - T_0) \frac{\partial \hat{T}}{\partial \hat{t}} \frac{d\hat{t}}{dt} = (T_{\text{end}} - T_0) \frac{\alpha}{L^2} \frac{\partial \hat{T}}{\partial \hat{t}}, \quad \frac{\partial q}{\partial x} = \frac{\bar{q}_0}{L} \frac{\partial \hat{q}}{\partial \hat{x}}. \quad (25)$$

Substituting into (10):

$$\rho c (T_{\text{end}} - T_0) \frac{\alpha}{L^2} \frac{\partial \hat{T}}{\partial \hat{t}} + \frac{\bar{q}_0}{L} \frac{\partial \hat{q}}{\partial \hat{x}} = 0. \quad (26)$$

Insert  $T_{\text{end}} - T_0 = \bar{q}_0 t_p / (\rho c L)$  from (23) into the first coefficient:

$$\rho c \cdot \frac{\bar{q}_0 t_p}{\rho c L} \cdot \frac{\alpha}{L^2} = \frac{\bar{q}_0 \alpha t_p}{L^3} = \frac{\bar{q}_0}{L} \cdot \frac{\alpha t_p}{L^2} = \frac{\bar{q}_0}{L} \hat{\tau}_\Delta. \quad (27)$$

So (26) becomes  $(\bar{q}_0 / L) [\hat{\tau}_\Delta \partial_{\hat{t}} \hat{T} + \partial_{\hat{x}} \hat{q}] = 0$ , and dividing by  $\bar{q}_0 / L \neq 0$  gives, dropping hats,

$$\boxed{\tau_\Delta \frac{\partial T}{\partial t} + \frac{\partial q}{\partial x} = 0} \quad (28)$$

**Derivation 4.2** (Constitutive law in dimensionless variables). Each term of (11) transforms as

$$\tau_q \frac{\partial q}{\partial t} = \tau_q \bar{q}_0 \frac{\alpha}{L^2} \frac{\partial \hat{q}}{\partial \hat{t}} = \bar{q}_0 \hat{\tau}_q \frac{\partial \hat{q}}{\partial \hat{t}}, \quad (29)$$

$$q = \bar{q}_0 \hat{q}, \quad (30)$$

$$\lambda \frac{\partial T}{\partial x} = \lambda (T_{\text{end}} - T_0) \frac{1}{L} \frac{\partial \hat{T}}{\partial \hat{x}} = \lambda \frac{\bar{q}_0 t_p}{\rho c L} \cdot \frac{1}{L} \frac{\partial \hat{T}}{\partial \hat{x}} = \bar{q}_0 \frac{\alpha t_p}{L^2} \frac{\partial \hat{T}}{\partial \hat{x}} = \bar{q}_0 \hat{\tau}_\Delta \frac{\partial \hat{T}}{\partial \hat{x}}, \quad (31)$$

$$\kappa^2 \frac{\partial^2 q}{\partial x^2} = \kappa^2 \frac{\bar{q}_0}{L^2} \frac{\partial^2 \hat{q}}{\partial \hat{x}^2} = \bar{q}_0 \hat{\kappa}^2 \frac{\partial^2 \hat{q}}{\partial \hat{x}^2}. \quad (32)$$

In (31) the step  $\lambda / (\rho c) = \alpha$  was used. Every term carries the common factor  $\bar{q}_0$ ; dividing it out and dropping hats,

$$\boxed{\tau_q \frac{\partial q}{\partial t} + q + \tau_\Delta \frac{\partial T}{\partial x} - \kappa^2 \frac{\partial^2 q}{\partial x^2} = 0} \quad (33)$$

---

**CHECK.** Note that the coefficient of  $\partial_x T$  in (33) is  $\tau_\Delta$ , not 1 and not  $\alpha$ . This follows from (31) and depends on the choice  $\hat{T} = (T - T_0) / (T_{\text{end}} - T_0)$  with  $T_{\text{end}}$  given by (23). A different temperature normalisation would change this coefficient. The appearance of the same  $\tau_\Delta$  in (28) and (33) is therefore not a coincidence but a consequence of the scaling; the program must use both.

---

**Derivation 4.3** (Boundary and initial conditions, dimensionless). With  $\hat{q} = q/\bar{q}_0$  and  $\bar{q}_0 = q_{\max}$  from (22), and  $t/t_p = \hat{t}/\hat{\tau}_\Delta$ , the pulse (15) becomes

$$q(0, t) = \begin{cases} 1 - \cos\left(\frac{2\pi t}{\tau_\Delta}\right), & 0 < t \leq \tau_\Delta, \\ 0, & t > \tau_\Delta, \end{cases} \quad q(1, t) = 0, \quad (34)$$

and the initial conditions are

$$T(x, 0) = 0, \quad q(x, 0) = 0, \quad \frac{\partial q}{\partial t}(x, 0) = 0, \quad (35)$$

the last by derivation 3.1. The domain is  $0 \leq x \leq 1$ .

**Derivation 4.4** (Dimensionless flux equation). Applying the same substitutions to (14), or eliminating  $T$  between (28) and (33),

$$\boxed{\tau_q \frac{\partial^2 q}{\partial t^2} + \frac{\partial q}{\partial t} = \frac{\partial^2 q}{\partial x^2} + \kappa^2 \frac{\partial^3 q}{\partial t \partial x^2}} \quad (36)$$

The coefficient of  $\partial_{xx}q$  is unity: the dimensionless diffusivity is  $\hat{\alpha} = \alpha t_p/L^2/\hat{\tau}_\Delta = 1$  by construction.

---

**RESULT.** The scaling sets  $\hat{\alpha} \equiv 1$ . The diffusivity is absorbed into the time variable and cannot be read from the shape of a dimensionless curve; it is recovered only through the physical time axis. This is essential when converting between physical and dimensionless parameters in the program, and was the source of a scaling error in earlier work that mixed  $\hat{\alpha} \equiv 1$  with the physical  $\alpha$  in the same expression.

---

Table 2: Dimensional to dimensionless mapping.  $\alpha$  is eliminated from the dimensionless equations and reappears only through  $\hat{t}$ .

| Dimensional | Characteristic scale   | Dimensionless                        | Equation       |
|-------------|------------------------|--------------------------------------|----------------|
| $t$         | $L^2/\alpha$           | $\hat{t} = \alpha t/L^2$             | (21)           |
| $x$         | $L$                    | $\hat{x} = x/L$                      | (21)           |
| $T - T_0$   | $T_{\text{end}} - T_0$ | $\hat{T}$                            | (21),(23)      |
| $q$         | $\bar{q}_0 = q_{\max}$ | $\hat{q}$                            | (21),(22)      |
| $t_p$       | $L^2/\alpha$           | $\hat{\tau}_\Delta = \alpha t_p/L^2$ | (24)           |
| $\tau_q$    | $L^2/\alpha$           | $\hat{\tau}_q = \alpha \tau_q/L^2$   | (24)           |
| $\kappa^2$  | $L^2$                  | $\hat{\kappa}^2 = \kappa^2/L^2$      | (24)           |
| $\alpha$    | itself                 | $\hat{\alpha} \equiv 1$              | derivation 4.4 |
| $B$         | —                      | invariant                            | section 6      |

## 5 Laplace-domain formulation and the propagation factor

**Definition 5.1** (Laplace transform). For a function  $f(x, t)$  with  $f(x, 0^-) = 0$ ,

$$\hat{f}(x, s) = \int_0^\infty f(x, t) e^{-st} dt, \quad \mathcal{L}[\partial_t f] = s\hat{f} - f(x, 0). \quad (37)$$

A prime denotes  $\partial/\partial x$  throughout this section.

**Derivation 5.1** (Transformed field equations). Apply (37) to the dimensionless energy balance (28) using  $T(x, 0) = 0$ :

$$\tau_\Delta [s\hat{T} - T(x, 0)] + \hat{q}' = 0 \implies \tau_\Delta s\hat{T} + \hat{q}' = 0. \quad (38)$$

Apply it to the constitutive law (33) using  $q(x, 0) = 0$ :

$$\tau_q [s\hat{q} - q(x, 0)] + \hat{q} + \tau_\Delta \hat{T}' - \kappa^2 \hat{q}'' = 0 \implies \tau_q s\hat{q} + \hat{q} + \tau_\Delta \hat{T}' - \kappa^2 \hat{q}'' = 0. \quad (39)$$

**Derivation 5.2** (Elimination of  $\hat{T}$  and the propagation factor). Differentiate (38) with respect to  $x$ :

$$\tau_{\Delta} s \hat{T}' + \hat{q}'' = 0 \implies \tau_{\Delta} \hat{T}' = -\frac{\hat{q}''}{s}. \quad (40)$$

Substitute (40) into (39):

$$\tau_q s \hat{q} + \hat{q} - \frac{\hat{q}''}{s} - \kappa^2 \hat{q}'' = 0. \quad (41)$$

Group the  $\hat{q}$  terms and the  $\hat{q}''$  terms:

$$(1 + \tau_q s) \hat{q} = \left( \frac{1}{s} + \kappa^2 \right) \hat{q}''. \quad (42)$$

Multiply both sides by  $s$  and divide by  $(1 + \kappa^2 s)$ , which is nonzero for  $s$  in the right half plane since  $\kappa^2 \geq 0$ :

$$\hat{q}'' = \frac{s(1 + \tau_q s)}{1 + \kappa^2 s} \hat{q}. \quad (43)$$

Restoring  $\alpha$  explicitly, so that the relation is valid in both the dimensional and the dimensionless reading (with  $\alpha = 1$  in the latter),

$$\hat{q}'' = m^2(s) \hat{q}, \quad m^2(s) = \frac{s(1 + \tau_q s)}{\alpha + \kappa^2 s}$$

(44)

**CHECK.** [Origin of every factor in (44)]

- the factor  $s$  in the numerator: the single time derivative in the energy balance (28), entering through (40);
- the factor  $(1 + \tau_q s)$ : the terms  $\tau_q \partial_t q + q$  of the constitutive law (33), i.e. relaxation plus the flux itself;
- the term  $\alpha$  in the denominator: the conduction term  $\lambda \partial_x T$ , which becomes  $\tau_{\Delta} \partial_x T$  in (33) and contributes the  $1/s$  of (42);
- the term  $\kappa^2 s$  in the denominator: the nonlocal term  $-\kappa^2 \partial_{xx} q$  of (33).

**Symbolic verification.** Check 1a of `check_symbolic.py` performs the elimination independently with `sympy.solve` and returns  $s(1 + \tau_q s)/(1 + \kappa^2 s)$ , matching (43) exactly. Check 1b confirms that (44) reduces to it at  $\alpha = 1$ . [NUMERICALLY COMPUTED HERE]

**CHECK.** [Dimensions of (44)]  $[\tau_q s] = s \cdot s^{-1} = 1$ , so  $(1 + \tau_q s)$  is dimensionally admissible.  $[\kappa^2 s] = m^2 \cdot s^{-1} = m^2 s^{-1} = [\alpha]$ , so  $(\alpha + \kappa^2 s)$  is admissible. Hence  $[m^2] = s^{-1}/m^2 s^{-1} = m^{-2}$ , and  $[m] = m^{-1}$ , correct for a quantity conjugate to  $x$ . Checks 2a–2c. [NUMERICALLY COMPUTED HERE]

**Definition 5.2** (Branch of  $m(s)$ ).  $m(s) = \sqrt{m^2(s)}$  is taken on the principal branch,  $\text{Re } m > 0$ . Since  $\alpha, \tau_q, \kappa^2 > 0$ , the zero of the numerator is at  $s = -1/\tau_q$  and the pole of the denominator at  $s = -\alpha/\kappa^2$ , both on the negative real axis. Therefore  $m^2(s)$  is analytic and non-vanishing in the open right half plane, and the principal branch is well defined there. This is the branch required by the inversion contour of section 21.

**Derivation 5.3** (Spatial solution and the observable). The general solution of (44) is  $\hat{q} = A \cosh(mx) + C \sinh(mx)$ . Imposing  $\hat{q}(0, s) = \hat{Q}_0(s)$  and  $\hat{q}(1, s) = 0$  and solving the two-by-two system gives the compact form

$$\hat{q}(x, s) = \hat{Q}_0(s) \frac{\sinh[m(1-x)]}{\sinh m}. \quad (45)$$

Recover the temperature from (38),  $\hat{T} = -\hat{q}'/(\tau_\Delta s)$ . Differentiating (45),  $\hat{q}' = -m\hat{Q}_0 \cosh[m(1-x)] / \sinh m$ , so

$$\boxed{\hat{T}(x, s) = \frac{m \hat{Q}_0(s) \cosh[m(1-x)]}{\tau_\Delta s \sinh m}} \quad (46)$$

and at the observed rear face  $x = 1$ , using  $\cosh 0 = 1$ ,

$$\boxed{\hat{T}(1, s) = \frac{m \hat{Q}_0(s)}{\tau_\Delta s \sinh m}} \quad (47)$$

---

**CHECK.** Checks 5c–5d of `check_symbolic.py` verify by substitution that (45) satisfies (44) and both boundary conditions  $\hat{q}(0) = \hat{Q}_0$ ,  $\hat{q}(1) = 0$ , and that  $-\hat{q}'/(\tau_\Delta s)$  equals (46). [NUMERICALLY COMPUTED HERE]

---

**Derivation 5.4** (Transform of the pulse). With  $\omega = 2\pi/\tau_\Delta$ , direct integration of the finite cosine pulse (34) gives

$$\begin{aligned} \hat{Q}_0(s) &= \int_0^{\tau_\Delta} \left[ 1 - \cos\left(\frac{2\pi u}{\tau_\Delta}\right) \right] e^{-su} du \\ &= \frac{\omega^2 (1 - e^{-s\tau_\Delta})}{s(s^2 + \omega^2)}. \end{aligned} \quad (48)$$

---

**CHECK.** [Removable singularities — numerically critical] The factor  $s^2 + \omega^2$  vanishes at  $s = \pm i\omega$ . At those points

$$1 - e^{-s\tau_\Delta} \Big|_{s=\pm i\omega} = 1 - e^{\mp 2\pi i} = 1 - 1 = 0, \quad (49)$$

so the numerator vanishes simultaneously and the apparent poles are *removable*:  $\hat{Q}_0$  is analytic on the imaginary axis. Checks 7a and 7b confirm (48) and (49) symbolically. [NUMERICALLY COMPUTED HERE]

**Consequence for the program.** Any numerical scheme that splits (48) into  $\omega^2/[s(s^2 + \omega^2)]$  times a shift operator reinstates genuine poles at  $s = \pm i\omega$  and will fail. This is quantified in section 21 and is a mandatory requirement in the program specification of section 23.

---



---

**RESULT.** The parameters  $\tau_q$  and  $\kappa^2$  enter the observable (47) *only* through  $m(s)$ . The pulse transform  $\hat{Q}_0$  depends on neither. This factorisation is what makes the analysis of section 9 exact and independent of the pulse shape.

---

## 6 Definition and interpretation of $B$

**Derivation 6.1** (Emergence of the ratio). The denominator of (44) is  $\alpha + \kappa^2 s$ . Factor out  $\alpha$ :

$$\alpha + \kappa^2 s = \alpha \left( 1 + \frac{\kappa^2}{\alpha} s \right) = \alpha \left( 1 + \frac{\kappa^2}{\alpha \tau_q} \tau_q s \right). \quad (50)$$

The numerator is  $s(1 + \tau_q s)$ . Both now contain the same grouping  $\tau_q s$ , and the only surviving free coefficient is the dimensionless combination

$$\boxed{B = \frac{\kappa^2}{\alpha \tau_q}} \quad (51)$$

so that

$$m^2(s) = \frac{s(1 + \tau_q s)}{\alpha(1 + B \tau_q s)}. \quad (52)$$

---

**RESULT.** [Mathematical meaning of  $B$ ] Equation (52) shows exactly what  $B$  is: the ratio of the pole location to the zero location of  $m^2$  on the negative real axis. The numerator vanishes at  $\tau_q s = -1$ ; the denominator vanishes at  $B\tau_q s = -1$ .  $B$  measures how far the nonlocal contribution shifts the pole relative to the relaxation zero. When  $B = 1$  the two coincide and cancel; this is the content of section 8.

---



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**CHECK.** [Dimensions]  $[B] = \frac{[\kappa^2]}{[\alpha][\tau_q]} = \frac{\text{m}^2}{\text{m}^2 \text{s}^{-1} \cdot \text{s}} = \frac{\text{m}^2}{\text{m}^2} = 1$ . Dimensionless. Check 4a. [NUMERICALLY COMPUTED HERE]

---

**Calculation 6.1** (Scale invariance: thickness does not change  $B$ ). Evaluate  $B$  in the dimensionless variables of (24), remembering  $\hat{\alpha} \equiv 1$ :

$$\hat{B} = \frac{\hat{\kappa}^2}{\hat{\alpha} \hat{\tau}_q} = \frac{\kappa^2/L^2}{1 \cdot (\alpha\tau_q/L^2)} = \frac{\kappa^2}{L^2} \cdot \frac{L^2}{\alpha\tau_q} = \frac{\kappa^2}{\alpha\tau_q} = B. \quad (53)$$

The thickness  $L$  cancels identically.

---

**RESULT.**  $B$  takes the same value in dimensional and dimensionless variables, and all thicknesses of a given material share one value of  $B$ . Within this 1-D model, specimen thickness is *not* a means of moving  $B$ . Check 4d verifies (53) symbolically. [NUMERICALLY COMPUTED HERE]

---

## 6.1 Reparameterisation

**Definition 6.1** (Two parameter vectors).

$$\boldsymbol{\theta} = (\alpha, \tau_q, \kappa^2), \quad \boldsymbol{\phi} = (\alpha, \tau_q, B). \quad (54)$$

**Derivation 6.2** (Forward and inverse maps). From (51), the forward map  $\boldsymbol{\theta} \mapsto \boldsymbol{\phi}$  is  $\alpha \mapsto \alpha$ ,  $\tau_q \mapsto \tau_q$ ,  $B = \kappa^2/(\alpha\tau_q)$ . Solving (51) for  $\kappa^2$  gives the inverse map

$$\boxed{\kappa^2 = \alpha \tau_q B} \quad (55)$$

**Calculation 6.2** (Jacobian of the transformation). With  $\boldsymbol{\theta} = (\alpha, \tau_q, \kappa^2)$  ordered as rows and  $\boldsymbol{\phi} = (\alpha, \tau_q, B)$  as columns, using (55),

$$\frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\phi}} = \begin{pmatrix} \frac{\partial \alpha}{\partial \alpha} & \frac{\partial \alpha}{\partial \tau_q} & \frac{\partial \alpha}{\partial B} \\ \frac{\partial \tau_q}{\partial \alpha} & \frac{\partial \tau_q}{\partial \tau_q} & \frac{\partial \tau_q}{\partial B} \\ \frac{\partial \kappa^2}{\partial \alpha} & \frac{\partial \kappa^2}{\partial \tau_q} & \frac{\partial \kappa^2}{\partial B} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \tau_q B & \alpha B & \alpha \tau_q \end{pmatrix}, \quad (56)$$

whose determinant, expanding along the first row, is

$$\det \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\phi}} = 1 \cdot (1 \cdot \alpha \tau_q - 0) = \alpha \tau_q. \quad (57)$$

---

**RESULT.**  $\det(\partial \boldsymbol{\theta} / \partial \boldsymbol{\phi}) = \alpha \tau_q \neq 0$  for all  $\alpha, \tau_q > 0$ . The reparameterisation is therefore a smooth bijection on the positive octant and creates or destroys no information anywhere, including at  $B = 1$ . Any degeneracy found in section 9 is a property of the map from parameters to observations, not an artefact of the coordinates. Checks 4b and 4c verify (55) and (57) symbolically. [NUMERICALLY COMPUTED HERE]

This change of variables is an elementary device used to organise the analysis. It is not presented as a result.

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## 7 Limiting cases

All three limits are taken in the *same* governing relation (44), so that the four models compared in section 16 differ only by constraints on  $(\tau_q, \kappa^2)$  and share one solver.

**Derivation 7.1** (Fourier limit). Set  $\tau_q \rightarrow 0$  and  $\kappa^2 \rightarrow 0$  in (44):

$$m^2(s) = \frac{s(1 + \tau_q s)}{\alpha + \kappa^2 s} \xrightarrow[\kappa^2 \rightarrow 0]{\tau_q \rightarrow 0} \frac{s(1 + 0)}{\alpha + 0} = \frac{s}{\alpha}. \quad (58)$$

Inverting  $\hat{q}'' = (s/\alpha)\hat{q}$  corresponds to  $\partial_t T = \alpha \partial_{xx} T$ , the diffusion equation.

**Interpretation.** One parameter,  $\alpha$ . Infinite propagation speed, purely diffusive rise. **Numerical check:** setting  $\tau_q, \kappa^2 \rightarrow 0$  in the solver reproduces the analytical Fourier solution to  $2.6 \times 10^{-6}$  [PREVIOUSLY VERIFIED].

**Derivation 7.2** (Maxwell–Cattaneo–Vernotte limit). Set  $\kappa^2 \rightarrow 0$  with  $\tau_q > 0$ :

$$m^2(s) \xrightarrow[\kappa^2 \rightarrow 0]{} \frac{s(1 + \tau_q s)}{\alpha} = \frac{s}{\alpha} + \frac{\tau_q s^2}{\alpha}. \quad (59)$$

The  $s^2$  term is the transform of  $\partial_{tt}$ , so this corresponds to  $\tau_q \partial_{tt} T + \partial_t T = \alpha \partial_{xx} T$ , a damped wave equation.

**Interpretation and wave speed.** As  $s \rightarrow \infty$ ,  $m^2 \sim \tau_q s^2/\alpha$ , so  $m \sim s\sqrt{\tau_q/\alpha}$  and the characteristic  $e^{-mx}$  propagates a front at finite speed

$$v = \sqrt{\alpha/\tau_q}. \quad (60)$$

Check 3d verifies  $\lim_{s \rightarrow \infty} m^2/s^2 = \tau_q/\alpha$  symbolically. [NUMERICALLY COMPUTED HERE] Two parameters,  $(\alpha, \tau_q)$ .

**Derivation 7.3** (Nyíri limit). Set  $\tau_q \rightarrow 0$  with  $\kappa^2 > 0$ :

$$m^2(s) \xrightarrow[\tau_q \rightarrow 0]{} \frac{s}{\alpha + \kappa^2 s}. \quad (61)$$

This retains the nonlocal term and discards relaxation.

**SOURCE.** The entropy-current formulation underlying this reduced form is due to Nyíri [7]; it is used here solely as the constraint  $\tau_q = 0$  on the same transfer function, because it isolates the mechanism that the Maxwell–Cattaneo–Vernotte limit omits. Two parameters,  $(\alpha, \kappa^2)$ .

**Derivation 7.4** (Full Guyer–Krumhansl model). No constraint:  $m^2(s) = s(1 + \tau_q s)/(\alpha + \kappa^2 s)$ , three parameters  $(\alpha, \tau_q, \kappa^2)$ .

---

**CHECK.** Dimensions of every limit: each of (58), (59) and (61) has dimension  $\text{m}^{-2}$ , since  $[s/\alpha] = \text{s}^{-1} \cdot \text{s} \cdot \text{m}^{-2} = \text{m}^{-2}$  and  $[\tau_q s^2/\alpha] = \text{s} \cdot \text{s}^{-2} \cdot \text{s} \cdot \text{m}^{-2} = \text{m}^{-2}$ . Checks 3a–3c verify all three limits symbolically. [NUMERICALLY COMPUTED HERE]

---

Table 3: The four models as constraints on one transfer function (44).  $k$  is the number of free parameters, used in the BIC penalty of section 16.

| Model   | Constraint                 | $m^2(s)$                                | $k$ |
|---------|----------------------------|-----------------------------------------|-----|
| Fourier | $\tau_q = 0, \kappa^2 = 0$ | $s/\alpha$                              | 1   |
| MCV     | $\kappa^2 = 0$             | $s(1 + \tau_q s)/\alpha$                | 2   |
| Nyíri   | $\tau_q = 0$               | $s/(\alpha + \kappa^2 s)$               | 2   |
| GK      | none                       | $s(1 + \tau_q s)/(\alpha + \kappa^2 s)$ | 3   |

MGT is excluded from this set: it requires a thermal-displacement variable that cannot be written as a constraint on (44), so including it would break the shared-solver protocol.

## 8 The Fourier-resonance condition

**Proposition 8.1** (Cancellation at  $B = 1$ ). For  $\alpha, \tau_q > 0$ , the condition  $B = 1$  is equivalent to  $\kappa^2 = \alpha\tau_q$ , and under it  $m^2(s)$  reduces identically to the Fourier propagation factor.

*Proof.* From (51),  $B = 1 \iff \kappa^2/(\alpha\tau_q) = 1 \iff \kappa^2 = \alpha\tau_q$ . Substitute  $\kappa^2 = \alpha\tau_q$  into the denominator of (44), one step at a time:

$$\alpha + \kappa^2 s = \alpha + (\alpha\tau_q) s \quad (62)$$

$$= \alpha(1 + \tau_q s). \quad (63)$$

Therefore

$$m^2(s)|_{B=1} = \frac{s(1 + \tau_q s)}{\alpha(1 + \tau_q s)} \quad (64)$$

$$= \frac{s}{\alpha} \cdot \frac{(1 + \tau_q s)}{(1 + \tau_q s)} \quad (65)$$

$$= \frac{s}{\alpha}, \quad (66)$$

the cancellation in (65) being legitimate for every  $s$  because  $\tau_q > 0$  implies  $(1 + \tau_q s) \neq 0$ . Comparing (66) with the Fourier limit (58) shows the two are identical.  $\square$

---

**CHECK.** Check 5a verifies the factorisation (63) ( $\alpha + \kappa^2 s$  with  $\kappa^2 = \alpha\tau_q$  factors to  $\alpha(1 + \tau_q s)$ ), and check 5b verifies  $m^2|_{B=1} - s/\alpha = 0$  exactly. [NUMERICALLY COMPUTED HERE]

---



---

**RESULT.** On the ray  $\kappa^2 = \alpha\tau_q$  the Guyer–Krumhansl propagation factor is *exactly* the Fourier one, for every  $s$ , and the relaxation time has disappeared from it entirely. The single factor  $(1 + \tau_q s)$  carried the whole dependence of  $m^2$  on  $\tau_q$ ; when it cancels,  $\tau_q$  leaves the problem.

---

**SOURCE.** The Fourier-resonance condition is not claimed as a discovery of this study. It is established in the literature. Fülöp, Kovács and Ván describe a hierarchical resonance at which the solutions of the coupled system may be identical to those of the single Fourier equation [10]. The validation anchor states it explicitly and names it, writing that the GK equation “is able to recover the solution of Fourier equation when  $\kappa^2/\tau = \alpha$  holds, called Fourier resonance” [1]. Fülöp et al. describe the Fourier solutions as being covered by those of the GK equation [11]. Fehér and Kovács use the same ratio as the lower limit of their fitting domain [8]. The calculations of section 9 concern the *consequences* of this known condition for the inverse problem.

## 9 Exact structural identifiability

**Definition 9.1** (Structural identifiability). Let  $\mathcal{M}(\theta)$  be the map from parameters to the noise-free observable  $\{T(1, t) : t > 0\}$ . A parameter or combination is structurally identifiable if  $\mathcal{M}(\theta_1) = \mathcal{M}(\theta_2)$  for all  $t$  implies equality of that parameter in  $\theta_1$  and  $\theta_2$ . Structural non-identifiability is a property of the model and the observation operator alone: no reduction of noise and no increase of sampling density can remove it.

### 9.1 Complete dependence of the forward solution

**Derivation 9.1** (The response on the resonance ray). By (46), the full dependence of the observable on  $(\tau_q, \kappa^2)$  is through  $m(s)$  alone;  $\hat{Q}_0$ ,  $\tau_\Delta$ ,  $s$  and  $x$  do not involve them. By proposition 8.1, on the ray

$$m|_{B=1} = \sqrt{\frac{s}{\alpha}}, \quad (67)$$

which contains no  $\tau_q$ . Substituting (67) into (46),

$$\hat{T}(x, s) \Big|_{B=1} = \frac{\sqrt{s/\alpha} \hat{Q}_0(s) \cosh \left[ \sqrt{s/\alpha} (1-x) \right]}{\tau_\Delta s \sinh \sqrt{s/\alpha}}. \quad (68)$$

Every symbol on the right is independent of  $\tau_q$ . Hence

$$\frac{\partial}{\partial \tau_q} \hat{T}(x, s) \Big|_{B=1} \equiv 0 \quad \forall s, \forall x \in [0, 1]. \quad (69)$$

The Laplace transform is injective on continuous functions, so the same holds for  $T(x, t)$  at every  $t$ .

---

**CHECK.** Check 5c confirms  $\partial_{\tau_q} m|_{\text{ray}} = 0$ , and check 5d differentiates the *full* expression (68) with respect to  $\tau_q$  at arbitrary  $x$  and returns exactly 0. [NUMERICALLY COMPUTED HERE]

---



---

**RESULT.** [Non-injectivity] Varying  $\tau_q$  while holding  $\kappa^2 = \alpha\tau_q$  traces the one-parameter family

$$\mathcal{R} = \{(\alpha, \tau_q, \alpha\tau_q) : \tau_q > 0\} \quad (70)$$

along which the observable is constant.  $\mathcal{M}$  is therefore not injective, and by definition 9.1 the pair  $(\tau_q, \kappa^2)$  is structurally non-identifiable on  $\mathcal{R}$ . This is not a coincidence of two particular solutions but an entire continuum of indistinguishable parameter sets.

---

## 9.2 Sensitivities

**Derivation 9.2** (Derivatives of  $m$  with respect to the parameters). Write  $m = (N/D)^{1/2}$  with  $N = s(1 + \tau_q s)$  and  $D = \alpha + \kappa^2 s$ . Then  $\partial_p m = \frac{1}{2} m^{-1} \partial_p (N/D)$ .

For  $p = \tau_q$ :  $\partial_{\tau_q} N = s \cdot s = s^2$  and  $\partial_{\tau_q} D = 0$ , so  $\partial_{\tau_q} (N/D) = s^2/D$  and

$$\frac{\partial m}{\partial \tau_q} = \frac{1}{2m} \cdot \frac{s^2}{\alpha + \kappa^2 s} = \frac{s^2}{2m(\alpha + \kappa^2 s)}. \quad (71)$$

For  $p = \kappa^2$ :  $\partial_{\kappa^2} N = 0$  and  $\partial_{\kappa^2} D = s$ , so  $\partial_{\kappa^2} (N/D) = -Ns/D^2 = -m^2 s/D$  and

$$\frac{\partial m}{\partial \kappa^2} = \frac{1}{2m} \cdot \left( -\frac{m^2 s}{\alpha + \kappa^2 s} \right) = -\frac{m s}{2(\alpha + \kappa^2 s)}. \quad (72)$$

---

**CHECK.** Checks 5e and 5f verify (71) and (72) against direct symbolic differentiation of  $m$ . [NUMERICALLY COMPUTED HERE]

---

**Derivation 9.3** (Sensitivity ratio). Because  $\tau_q$  and  $\kappa^2$  enter  $\hat{T}$  only through  $m$ , the chain rule gives  $\partial_p \hat{T} = (\partial_m \hat{T})(\partial_p m)$  for  $p \in \{\tau_q, \kappa^2\}$ . Wherever  $\partial_m \hat{T} \neq 0$  the common factor cancels in a ratio:

$$\frac{\partial \hat{T} / \partial \tau_q}{\partial \hat{T} / \partial \kappa^2} = \frac{\partial m / \partial \tau_q}{\partial m / \partial \kappa^2}. \quad (73)$$

Dividing (71) by (72), the factor  $2(\alpha + \kappa^2 s)$  cancels:

$$\frac{\partial m / \partial \tau_q}{\partial m / \partial \kappa^2} = \frac{s^2/m}{-m s} = -\frac{s}{m^2}. \quad (74)$$

This holds for *all* parameter values. Now evaluate on the ray, where  $m^2 = s/\alpha$  by proposition 8.1:

$$\boxed{\frac{\partial \hat{T} / \partial \tau_q}{\partial \hat{T} / \partial \kappa^2} \Big|_{B=1} = -\frac{s}{s/\alpha} = -\alpha} \quad (75)$$



---

**RESULT.** The ratio (75) is a *constant*: independent of  $s$ , hence independent of  $t$  after inversion; independent of  $x$ ; and independent of the pulse, since  $\hat{Q}_0$  cancelled in (73). Therefore in the time domain, pointwise,

$$\frac{\partial T}{\partial \tau_q} = -\alpha \frac{\partial T}{\partial \kappa^2}. \quad (76)$$

The two sensitivity fields are exactly anti-parallel. Checks 5g and 5h verify (74) and (75).  
[NUMERICALLY COMPUTED HERE]

---

### 9.3 Jacobian rank, Fisher singularity, correlation

**Definition 9.2** (Sensitivity matrix). For observations  $y_i = T(1, t_i)$ ,  $i = 1, \dots, n$ , and parameters  $\theta_j$ ,

$$J_{ij} = \frac{\partial y_i}{\partial \theta_j}. \quad (77)$$

**Derivation 9.4** (Rank deficiency of the  $(\tau_q, \kappa^2)$  block). Restrict to the two columns  $j \in \{\tau_q, \kappa^2\}$  and evaluate on the ray. By (76), for every observation index  $i$ ,

$$J_{i, \tau_q} = -\alpha J_{i, \kappa^2}, \quad (78)$$

that is, the first column is a constant multiple of the second:

$$J_{:, \tau_q} = -\alpha J_{:, \kappa^2}. \quad (79)$$

Two linearly dependent columns give

$$\text{rank} \begin{bmatrix} J_{:, \tau_q} & J_{:, \kappa^2} \end{bmatrix} = 1, \quad (80)$$

and the null direction follows from  $J_{:, \tau_q} \cdot 1 + J_{:, \kappa^2} \cdot \alpha = -\alpha J_{:, \kappa^2} + \alpha J_{:, \kappa^2} = \mathbf{0}$ , i.e.

$$\begin{bmatrix} J_{:, \tau_q} & J_{:, \kappa^2} \end{bmatrix} \begin{pmatrix} 1 \\ \alpha \end{pmatrix} = \mathbf{0}. \quad (81)$$

---

**RESULT.** [Geometric meaning of the null vector] The direction  $(1, \alpha)$  in the  $(\tau_q, \kappa^2)$  plane is precisely the tangent to the ray  $\kappa^2 = \alpha \tau_q$ , since along it  $d\kappa^2/d\tau_q = \alpha$ . The flat direction of the objective function *is* the resonance ray.

---

**Derivation 9.5** (Fisher information and its singularity). Assume additive independent Gaussian noise of constant variance  $\sigma^2$ ,  $y_i^{\text{obs}} = y_i(\boldsymbol{\theta}) + \varepsilon_i$ ,  $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ . The log-likelihood is

$$\ln \mathcal{L} = \text{const} - \frac{1}{2\sigma^2} \sum_{i=1}^n \left[ y_i^{\text{obs}} - y_i(\boldsymbol{\theta}) \right]^2, \quad (82)$$

and the Fisher information matrix is

$$F_{jk} = \mathbb{E} \left[ \frac{\partial \ln \mathcal{L}}{\partial \theta_j} \frac{\partial \ln \mathcal{L}}{\partial \theta_k} \right] = \frac{1}{\sigma^2} \sum_{i=1}^n \frac{\partial y_i}{\partial \theta_j} \frac{\partial y_i}{\partial \theta_k} \quad \Longleftrightarrow \quad F = \frac{J^\top J}{\sigma^2}. \quad (83)$$

On the ray, substituting (79) into the two-by-two block,

$$F^{(\tau_q, \kappa^2)} = \frac{1}{\sigma^2} \begin{pmatrix} \alpha^2 \|J_{:, \kappa^2}\|^2 & -\alpha \|J_{:, \kappa^2}\|^2 \\ -\alpha \|J_{:, \kappa^2}\|^2 & \|J_{:, \kappa^2}\|^2 \end{pmatrix} = \frac{\|J_{:, \kappa^2}\|^2}{\sigma^2} \begin{pmatrix} \alpha^2 & -\alpha \\ -\alpha & 1 \end{pmatrix}, \quad (84)$$

whose determinant is

$$\det F^{(\tau_q, \kappa^2)} = \left( \frac{\|J_{:, \kappa^2}\|^2}{\sigma^2} \right)^2 [\alpha^2 \cdot 1 - (-\alpha)(-\alpha)] = \left( \frac{\|J_{:, \kappa^2}\|^2}{\sigma^2} \right)^2 [\alpha^2 - \alpha^2] = 0. \quad (85)$$

The implied parameter correlation is

$$\text{corr}(\tau_q, \kappa^2) = \frac{F_{12}}{\sqrt{F_{11}F_{22}}} = \frac{-\alpha}{\sqrt{\alpha^2 \cdot 1}} = -1. \quad (86)$$

**Proposition 9.1** (Structural non-identifiability at  $B = 1$ ). For the system of section 1 with  $\kappa^2 = \alpha\tau_q$ :

- (i)  $T(x, t)$  is independent of  $\tau_q$  at every  $x \in [0, 1]$  and every  $t$ , for any pulse;
- (ii) the family  $\mathcal{R}$  of (70) produces identical observations, so  $(\tau_q, \kappa^2)$  are not separately structurally identifiable;
- (iii)  $\partial T / \partial \tau_q = -\alpha \partial T / \partial \kappa^2$  exactly, the sensitivity matrix has rank one with null direction  $(1, \alpha)$ , the Fisher matrix is singular, and  $\text{corr}(\tau_q, \kappa^2) = -1$ .

*Proof.* (i) is derivation 9.1, equation (69). (ii) follows from (i) by definition 9.1. (iii) is (76), (80), (81), (85) and (86).  $\square$

---

**CHECK.** [Independent symbolic verification of the whole proposition] Checks 6a–6d of `check_symbolic.py` construct the Jacobian row  $[g \partial_{\tau_q} m, g \partial_{\kappa^2} m]$  with  $g = \partial_m T$  symbolic and nonzero, evaluate on  $\kappa^2 = \alpha\tau_q$ , and confirm: the entry ratio is  $-\alpha$ ;  $(1, \alpha)$  annihilates the row;  $\det(J^\top J) = 0$ ; the product of the two entries is negative, giving correlation  $-1$ .

*Technical note recorded for reproducibility.* Sympy does not automatically reduce  $\sqrt{s^2\tau_q^2 + 2s\tau_q + 1}$  to  $(1 + s\tau_q)$ . The polynomial factors exactly as  $(1 + s\tau_q)^2$  and both factors are positive for  $s, \tau_q > 0$ , so the reduction is valid; the check script applies it explicitly. Numerical spot checks confirm the difference is zero to machine precision. Without this reduction the checks return unsimplified expressions rather than 0; this is a limitation of the algebra system, not of the result. [NUMERICALLY COMPUTED HERE]

---



---

**RESULT.** [What is *not* claimed] proposition 9.1 concerns the pair  $(\tau_q, \kappa^2)$  only. It does *not* state that GK parameters are unidentifiable in general. By (68) the response on the ray is a Fourier response with diffusivity  $\alpha$ , so  $\alpha$  remains determined by the physical time axis, and the statement  $B = 1$  is itself an identified fact. See section 12.

---

## 10 Numerical proof that the degeneracy is exact

The purpose of this section is to exclude the possibility that proposition 9.1 is an artefact of algebraic manipulation, and to exclude discretisation, interpolation, inversion and round-off as explanations.

**Calculation 10.1** (Setup). **Parameter setup.** Three parameter sets on the ray, with  $\tau_q$  spanning two orders of magnitude and  $\kappa^2$  slaved to it:

$$\tau_q \in \{3 \times 10^{-3}, 3 \times 10^{-2}, 3 \times 10^{-1}\}, \quad \kappa^2 = \alpha\tau_q \quad (B = 1), \quad \alpha = 1. \quad (87)$$

**Control.** The identical three  $\tau_q$  values with  $\kappa^2 = 1.05\alpha\tau_q$ , i.e.  $B = 1.05$ , a 5% departure from resonance.

**Observation.** Rear face  $x = 1$ , on the grid  $t \in [0.005, 1.0]$  with 400 points,  $\tau_\Delta = 0.04$ .

**Comparison metric.** The pointwise spread across the three curves, maximised over the grid:

$$S = \max_t \left[ \max_j T_j(t) - \min_j T_j(t) \right], \quad j = 1, 2, 3. \quad (88)$$

**Expected result.** If proposition 9.1 holds,  $S$  on the ray is limited only by the numerical noise floor of the solver, whereas the control must show a spread of order the physical difference between  $B = 1$  and  $B = 1.05$ .

---

**RESULT.** [Double precision, certified convolution solver]

$$S_{\text{ray}} = 1.261 \times 10^{-11}, \quad [\text{NUMERICALLY COMPUTED HERE}] \quad (89)$$

$$S_{\text{control}} = 2.595 \times 10^{-2}, \quad [\text{NUMERICALLY COMPUTED HERE}] \quad (90)$$

$$S_{\text{control}}/S_{\text{ray}} = 2.06 \times 10^9. \quad [\text{NUMERICALLY COMPUTED HERE}] \quad (91)$$

Script `validation/calc/check_numerical.py`, checks 7a–7c.

---

**Calculation 10.2** (Independent arbitrary-precision confirmation). To exclude double-precision round-off entirely, the same test was repeated in 50-digit arithmetic using an independent implementation (`mpmath`, convolution of the kernel with the pulse, Talbot inversion at degree 30), at the single probe  $t = 0.5$ :

$$S_{\text{ray}}^{(50\text{-digit})} = 7.733 \times 10^{-30}. \quad [\text{NUMERICALLY COMPUTED HERE}] \quad (92)$$

Script `validation/calc/check_numerical.py`, check 7d.

---

**CHECK.** The spread falls from  $1.3 \times 10^{-11}$  in double precision to  $7.7 \times 10^{-30}$  at 50 digits, i.e. it tracks the working precision rather than settling at a fixed nonzero value. That is the signature of an exactly zero quantity computed approximately, not of a small but genuine difference. The control at  $B = 1.05$  does not collapse and is nine orders of magnitude larger, confirming that the effect is specific to the ray and is not a consequence of the solver, the grid or the observation point.

---

**RESULT.** Ruled out as explanations: discretisation, interpolation, Laplace inversion, finite precision and coding error. The degeneracy is exact, consistent with proposition 9.1.

---

## 11 Sensitivity matrix and Fisher information

**Definition 11.1** (Observation model).

$$y_i^{\text{obs}} = T(1, t_i; \boldsymbol{\theta}) + \varepsilon_i, \quad \varepsilon_i \text{ i.i.d. } \mathcal{N}(0, \sigma^2), \quad i = 1, \dots, n. \quad (93)$$

The noise level is specified as a fraction  $\eta$  of the temperature range:

$$\sigma = \eta \left[ \max_i T(1, t_i) - \min_i T(1, t_i) \right]. \quad (94)$$

**Definition 11.2** (Log-parameterisation). Estimation is performed in  $\ln \theta_j$ , which enforces positivity and makes the three parameters comparably scaled. The chain rule gives the log-Jacobian

$$\tilde{J}_{ij} = \frac{\partial y_i}{\partial \ln \theta_j} = \theta_j \frac{\partial y_i}{\partial \theta_j} = \theta_j J_{ij} \quad (\text{no sum}), \quad (95)$$

so standard errors obtained from  $\tilde{F} = \tilde{J}^\top \tilde{J} / \sigma^2$  are *relative* standard errors, reported below as percentages.

**Derivation 11.1** (Analytic sensitivities used by the program). Finite differencing of the time-domain solver is not used anywhere in the statistics, because the solver noise floor contaminates the Jacobian. Instead, writing the rear-face kernel of (144) below as  $\hat{K} = m/(\tau_\Delta s \sinh m)$ , differentiate with respect to  $m$  using the quotient rule:

$$\frac{\partial \hat{K}}{\partial m} = \frac{1}{\tau_\Delta s} \cdot \frac{1 \cdot \sinh m - m \cosh m}{\sinh^2 m} \quad (96)$$

$$= \frac{1}{\tau_\Delta s \sinh m} \left( 1 - m \frac{\cosh m}{\sinh m} \right) \quad (97)$$

$$= \frac{1}{\tau_\Delta s \sinh m} (1 - m \coth m). \quad (98)$$

Combining with (71) and (72),

$$\frac{\partial \hat{K}}{\partial \tau_q} = \frac{\partial \hat{K}}{\partial m} \cdot \frac{s^2}{2m(\alpha + \kappa^2 s)}, \quad \frac{\partial \hat{K}}{\partial \kappa^2} = \frac{\partial \hat{K}}{\partial m} \cdot \left( -\frac{ms}{2(\alpha + \kappa^2 s)} \right). \quad (99)$$

These are inverted and convolved by exactly the same procedure as the forward solution.

---

**CHECK.** Check 4c of the earlier forensic suite verifies (98) symbolically. The assembled analytic Jacobian was compared against central differences of the forward solver: maximum relative deviation  $5.3 \times 10^{-7}$  [PREVIOUSLY VERIFIED]. Because  $\alpha$  rescales the time axis and the pulse length rather than entering the kernel as a parameter, its sensitivity is obtained by a single central difference in  $\alpha$ ;  $\tau_q$  and  $\kappa^2$ , whose identifiability is the subject of this study, never use finite differences.

---

**Derivation 11.2** (Why  $B = 1$  makes  $F$  singular, algebraically). This restates derivation 9.5 in the form used by the program. The two columns of  $\tilde{J}$  corresponding to  $(\tau_q, \kappa^2)$  are, on the ray,  $\tilde{J}_{:, \tau_q} = \tau_q J_{:, \tau_q}$  and  $\tilde{J}_{:, \kappa^2} = \kappa^2 J_{:, \kappa^2} = \alpha \tau_q J_{:, \kappa^2}$ . Using  $J_{:, \tau_q} = -\alpha J_{:, \kappa^2}$ ,

$$\tilde{J}_{:, \tau_q} = -\alpha \tau_q J_{:, \kappa^2} = -\tilde{J}_{:, \kappa^2}. \quad (100)$$

In log-parameters the two columns are exactly opposite, so  $\tilde{J}_{:, \tau_q} + \tilde{J}_{:, \kappa^2} = \mathbf{0}$ : the null direction in log coordinates is  $(1, 1)$ , i.e. the direction along which  $\tau_q$  and  $\kappa^2$  are scaled by the same factor, which is precisely motion along the ray.

---

**RESULT.** The algebraic proof of singularity precedes and is independent of any numerical condition number. The condition numbers reported in section 13 are a consequence, not the evidence.

---

## 12 Reparameterisation: what remains determined

**Derivation 12.1** (Sensitivity in the  $\phi$  coordinates). Using  $\phi = (\alpha, \tau_q, B)$  and  $\kappa^2 = \alpha \tau_q B$ , the derivative of the observable with respect to  $\tau_q$  at fixed  $\alpha$  and  $B$  is obtained by the chain rule, noting  $\partial \kappa^2 / \partial \tau_q|_{\alpha, B} = \alpha B$ :

$$\left. \frac{\partial T}{\partial \tau_q} \right|_{\alpha, B} = \left. \frac{\partial T}{\partial \tau_q} \right|_{\alpha, \kappa^2} + \left. \frac{\partial T}{\partial \kappa^2} \cdot \frac{\partial \kappa^2}{\partial \tau_q} \right|_{\alpha, B} = \left. \frac{\partial T}{\partial \tau_q} \right|_{\alpha, \kappa^2} + \alpha B \left. \frac{\partial T}{\partial \kappa^2} \right|_{\alpha, B}. \quad (101)$$

On the resonance ray  $B = 1$ , and substituting (76),

$$\left. \frac{\partial T}{\partial \tau_q} \right|_{\alpha, B=1} = -\alpha \left. \frac{\partial T}{\partial \kappa^2} \right|_{\alpha, B=1} + \alpha \cdot 1 \cdot \left. \frac{\partial T}{\partial \kappa^2} \right|_{\alpha, B=1} = 0. \quad (102)$$

---

**CHECK.** Check 6b verifies  $\partial_{\tau_q} \kappa^2|_{\alpha, B} = \alpha B$  and that the bracket in (101) vanishes on the ray. [NUMERICALLY COMPUTED HERE]

---

**RESULT.** [Localisation of the degeneracy] In the coordinates  $\phi = (\alpha, B, \tau_q)$  the degeneracy is confined to the single coordinate  $\tau_q$ : the derivative with respect to  $\tau_q$  at fixed  $(\alpha, B)$  vanishes identically at  $B = 1$ , while the derivatives with respect to  $\alpha$  and  $B$  do not. The information carried by the data is carried by  $(\alpha, B)$ ; it is the resolution of that information into  $\tau_q$  and  $\kappa^2$  separately that fails.

---

## 12.1 Numerical evidence, kept separate from the theorem

The following is a numerical calculation and is *not* part of the proof of proposition 9.1. It quantifies how well  $(\alpha, B)$  are determined for one specific configuration.

**Calculation 12.1** (Standard errors in both parameterisations). Configuration as in section 13:  $\tau_q = 0.03$ ,  $\tau_\Delta = 0.04$ ,  $n = 80$  points on  $t \in [0.02, 1.2]$ ,  $\eta = 2\%$ , log-parameters, analytic Jacobian. The standard error of  $B$  is obtained from the delta method: with  $\ln B = \ln \kappa^2 - \ln \alpha - \ln \tau_q$  the gradient in log coordinates is  $g = (-1, -1, 1)$  and

$$\text{s.e.}(\ln B) = \sqrt{g^\top \tilde{F}^{-1} g}. \quad (103)$$

Table 4: Standard errors in the two parameterisations. All values [NUMERICALLY COMPUTED HERE], script `validation/audit/reident.py` and `check_numerical.py`. Configuration-specific.

| $B$         | s.e. $\tau_q$ [%] | s.e. $\kappa^2$ [%] | s.e. $\alpha$ [%] | s.e. $B$ [%] |
|-------------|-------------------|---------------------|-------------------|--------------|
| 0.50        | 12.37             | 21.56               | 0.94              | 9.56         |
| 0.90        | 98.75             | 107.3               | 1.04              | 9.03         |
| 0.98        | 526.6             | 535.0               | 1.06              | 8.90         |
| <b>1.00</b> | <b>divergent</b>  | <b>divergent</b>    | <b>0.94</b>       | <b>4.00</b>  |
| 1.10        | 114.7             | 106.5               | 1.08              | 8.69         |
| 1.28        | 45.64             | 37.80               | 1.11              | 8.39         |
| 5.00        | 8.12              | 3.72                | 1.49              | 5.68         |

---

**RESULT.**  $\alpha$  is determined to about 1% and  $B$  to between 4% and 10% across the whole range *including exactly at resonance*, while  $\tau_q$  and  $\kappa^2$  individually diverge. The data are not uninformative at  $B = 1$ ; the split is what fails. This is a numerical result for the stated configuration and does not extend the structural theorem.

---

## 13 Practical identifiability

**Definition 13.1** (Practical identifiability). Recoverability of a parameter to a stated precision from finite, noisy data. Unlike definition 9.1 it depends on the sampling design, the noise level and the estimator, and it degrades continuously rather than switching on at a point.

**Definition 13.2** (Reference configuration). Every number in this section and in sections 14 to 16 refers to:

$$\text{true parameters: } \alpha = 1, \quad \tau_q = 0.03, \quad \kappa^2 = B \alpha \tau_q, \quad (104)$$

$$\text{pulse: } \tau_\Delta = 0.04, \quad (105)$$

$$\text{sampling: } n = 80 \text{ points, uniform on } t \in [0.02, 1.2], \quad (106)$$

$$\text{noise: } \eta = 2\% \text{ of the temperature range,} \quad (107)$$

$$\text{estimator: Levenberg–Marquardt on } \ln \boldsymbol{\theta} \text{ with analytic Jacobian,} \quad (108)$$

$$\text{working units: } L = 1, \quad t_p = 0.04. \quad (109)$$

**Definition 13.3** (Objective function).

$$\text{SSE}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[ y_i^{\text{obs}} - T(1, t_i; \boldsymbol{\theta}) \right]^2, \quad (110)$$

minimised over  $\ln \boldsymbol{\theta}$ . Under (93) this is the maximum likelihood estimator.

**Calculation 13.1** (Fisher standard errors versus  $B$ ). For each  $B$ : build  $\tilde{J}$  analytically, form  $\tilde{F} = \tilde{J}^\top \tilde{J} / \sigma^2$ , invert, take  $\sqrt{\text{diag}}$ . Results in table 5.

Table 5: Practical identifiability versus  $B$  for the configuration of definition 13.2. All values [NUMERICALLY COMPUTED HERE]. Fisher from `reident.py`; Monte Carlo from `robust_mc.py` (200 trials, spread reported as interquartile range divided by 1.349, because the sampling distribution is heavy tailed near resonance).

| $B$         | $\text{cond}(\tilde{F})$               | $\text{corr}(\tau_q, \kappa^2)$ | Fisher s.e. $\tau_q$ [%] | MC s.e. $\tau_q$ [%] | MC $\tau_q$ 5–95% | profile factor   |
|-------------|----------------------------------------|---------------------------------|--------------------------|----------------------|-------------------|------------------|
| 0.10        | $3.6 \times 10^2$                      | +0.767                          | 2.05                     | —                    | —                 | —                |
| 0.30        | $4.8 \times 10^2$                      | +0.926                          | 5.83                     | —                    | —                 | —                |
| 0.50        | $9.3 \times 10^2$                      | +0.974                          | 12.37                    | 12.4                 | 0.84–1.23         | 1.4              |
| 0.70        | $2.7 \times 10^3$                      | +0.993                          | 27.06                    | —                    | —                 | —                |
| 0.90        | $2.5 \times 10^4$                      | +0.999                          | 98.75                    | 103.9                | 0.09–5.88         | 35.5             |
| 0.98        | $6.4 \times 10^5$                      | +1.000                          | 526.6                    | —                    | —                 | —                |
| <b>1.00</b> | <b><math>1.4 \times 10^{16}</math></b> | <b>singular</b>                 | <b>divergent</b>         | unbounded            | unbounded         | <b>unbounded</b> |
| 1.02        | $6.5 \times 10^5$                      | +1.000                          | 542.5                    | —                    | —                 | —                |
| 1.10        | $2.7 \times 10^4$                      | +0.999                          | 114.7                    | —                    | —                 | —                |
| 1.28        | $3.5 \times 10^3$                      | +0.996                          | 45.64                    | 48.9                 | 0.35–1.85         | 3.5              |
| 1.50        | $1.2 \times 10^3$                      | +0.987                          | 28.49                    | —                    | —                 | —                |
| 2.00        | $3.2 \times 10^2$                      | +0.957                          | 17.19                    | —                    | —                 | —                |
| 2.41        | $1.8 \times 10^2$                      | +0.927                          | 13.73                    | 13.3                 | 0.79–1.20         | —                |
| 3.00        | $9.8 \times 10^1$                      | +0.882                          | 11.17                    | —                    | —                 | —                |
| 5.00        | $3.6 \times 10^1$                      | +0.757                          | 8.12                     | 7.2                  | 0.88–1.12         | 1.3              |

**CHECK.** Checks 8-0.5, 8-1.28, 8-2.41, 8-5.0 reproduce the Fisher column to the quoted precision; check 8e confirms  $\text{cond}(\tilde{F}) = 1.4 \times 10^{16}$  at  $B = 1 + 10^{-7}$ . [NUMERICALLY COMPUTED HERE]

### 13.1 Derivation of the band boundaries

**Calculation 13.2** (Threshold definition and root-finding). Define  $\text{se}_{\tau_q}(B)$  as the Fisher relative standard error of  $\tau_q$  from calculation 13.1. For a target  $\Theta$  the band is the set where the target is *not* met:

$$\mathcal{B}(\Theta) = \{B : \text{se}_{\tau_q}(B) > \Theta\}. \quad (111)$$

Since  $\text{se}_{\tau_q}$  increases monotonically towards  $B = 1$  from either side,  $\mathcal{B}(\Theta)$  is an interval  $[B_-, B_+]$  whose endpoints solve  $\text{se}_{\tau_q}(B) = \Theta$ . Each endpoint is located by 50 bisection steps, bracketing from  $B = 1$  outwards.

---

**RESULT.** [Bands for the configuration of definition 13.2]

$$\text{se}_{\tau_q} > 10\% : \quad B \in [0.441, 3.464], \quad \text{[NUMERICALLY COMPUTED HERE]} \quad (112)$$

$$\text{se}_{\tau_q} > 20\% : \quad B \in [0.628, 1.804], \quad \text{[NUMERICALLY COMPUTED HERE]} \quad (113)$$

$$\text{se}_{\tau_q} > 50\% : \quad B \in [0.817, 1.252], \quad \text{[NUMERICALLY COMPUTED HERE]} \quad (114)$$

$$\text{se}_{\tau_q} > 100\% : \quad B \in [0.901, 1.116]. \quad \text{[NUMERICALLY COMPUTED HERE]} \quad (115)$$

Checks 8f and 8g reproduce the endpoints of (113) to  $\pm 0.002$ . The complementary design criterion, the region where the target *is* met, is  $B \leq B_-$  or  $B \geq B_+$ .

---

**CHECK.** [Scope of these numbers — mandatory qualification] (112)–(115) are properties of the configuration of definition 13.2. They are *not* universal physical constants and would move under a different sampling scheme, noise level or pulse length. They must be recomputed for any materially different design; the program specification of section 23 therefore exposes  $n$ ,  $\eta$ ,  $t_{\min}$ ,  $t_{\max}$  and  $\tau_{\Delta}$  as inputs. Only the singularity at  $B = 1$  is configuration-independent, because it is structural (proposition 9.1).

---

## 14 Profile likelihood

**Definition 14.1** (Profile of the objective). Fix  $\tau_q$  at a trial value  $\tau$  and re-optimize the remaining parameters:

$$\text{SSE}_{\text{prof}}(\tau) = \min_{\alpha, \kappa^2} \text{SSE}(\alpha, \tau, \kappa^2). \quad (116)$$

The profile likelihood ratio statistic is

$$\Delta(\tau) = \frac{\text{SSE}_{\text{prof}}(\tau) - \min_{\tau'} \text{SSE}_{\text{prof}}(\tau')}{\sigma^2}, \quad (117)$$

which under (93) is asymptotically  $\chi^2$  with one degree of freedom. The 95% confidence set is

$$\left\{ \tau : \Delta(\tau) \leq \chi_{1,0.95}^2 \right\}, \quad \chi_{1,0.95}^2 = 3.841. \quad (118)$$

**Why this is a stronger diagnostic than Fisher curvature.** The Fisher standard error is the curvature of SSE at the optimum and therefore assumes the objective is well approximated by a quadratic. Near a degeneracy the objective develops a long flat valley and that approximation fails; the Fisher error can then be finite while the true confidence set is unbounded. The profile makes no quadratic assumption: it evaluates the objective along the  $\tau_q$  direction with all other parameters free, so it reports the actual extent of the valley.

**Calculation 14.1** (Procedure). Grid: 53 values of  $\tau$  logarithmically spaced over  $\tau_q \times [10^{-1.3}, 10^{+1.3}]$ , i.e. a factor of  $\approx 400$  either side. At each node, (116) is solved by Levenberg–Marquardt on  $(\ln \alpha, \ln \kappa^2)$  with two starting points to reduce the risk of a local minimum. One noise realisation per  $B$ , seed fixed for reproducibility. Script `validation/audit/run_profile.py`.

Table 6: Profile-likelihood 95% intervals for  $\tau_q$ . All values [NUMERICALLY COMPUTED HERE]. Configuration of definition 13.2.

| $B$         | 95% interval for $\tau_q$ | As multiple of truth | Width factor             |
|-------------|---------------------------|----------------------|--------------------------|
| 0.50        | [0.0267, 0.0378]          | [0.89, 1.26]         | 1.4                      |
| 0.90        | [0.0015, 0.0534]          | [0.05, 1.78]         | 35.5                     |
| <b>1.00</b> | [0.0015, 0.5986]          | [0.05, 19.95]        | <b>398, grid-limited</b> |
| 1.28        | [0.0238, 0.0846]          | [0.79, 2.82]         | 3.5                      |
| 5.00        | [0.0337, 0.0424]          | [1.12, 1.41]         | 1.3                      |

**RESULT.** At  $B = 1$  the profile does not cross the threshold anywhere on a grid spanning a factor of 400: the interval is bounded only by the grid, so  $\tau_q$  is *unbounded* by the data, exactly as proposition 9.1 requires. The widths at the other  $B$  values increase monotonically towards resonance and are consistent with the Fisher column of table 5 where the quadratic approximation is valid.

The value at  $B = 1.28$ , a factor of 3.5, is the conditioning of a real calibration (see section 17) and brackets the factor 1.51 to 3.50 by which two published evaluations of the same specimens disagree.

## 15 Monte Carlo parameter recovery

**Calculation 15.1** (Procedure). (1) **Truth.**  $\theta^* = (1, 0.03, B \cdot 0.03)$ .

- (2) **Noise generation.** Compute  $y^* = T(1, t_i; \theta^*)$ ; set  $\sigma$  by (94) with  $\eta = 0.02$ ; draw  $y^{\text{obs}} = y^* + \mathcal{N}(0, \sigma^2)$  independently per point. Generator: `numpy PCG64`, seeds  $7000 + i$  (fixed, so the study is exactly reproducible).
- (3) **Trials.** 200 per value of  $B$ .
- (4) **Optimizer.** Levenberg–Marquardt (`scipy least_squares, method='lm'`) on  $\ln \theta$ , `xtol=ftol=10-13`, with the analytic Jacobian of derivation 11.1.
- (5) **Initial condition.** Started at the true parameters. This is the favourable case and is stated as such.
- (6) **Bounds.** None; positivity is enforced by the log-parameterisation.
- (7) **Summary statistic.** Because the sampling distribution is heavy tailed near resonance, the spread is reported as the interquartile range divided by 1.349 (the Gaussian-consistent robust estimator), alongside the 5–95 percentile range.

Script `validation/audit/robust_mc.py`.

Table 7: Monte Carlo recovery, 200 trials. All values [NUMERICALLY COMPUTED HERE]. Comparison column from table 5.

| $B$  | MC s.e. $\tau_q$ [%] | Fisher s.e. $\tau_q$ [%] | $\alpha$ 5–95% | $B$ 5–95%      |
|------|----------------------|--------------------------|----------------|----------------|
| 0.50 | 12.4                 | 12.37                    | [0.983, 1.017] | [0.842, 1.149] |
| 0.90 | 103.9                | 98.75                    | [0.982, 1.026] | [0.064, 1.070] |
| 1.00 | unbounded            | divergent                | unbounded      | unbounded      |
| 1.28 | 48.9                 | 45.64                    | [0.980, 1.019] | [0.906, 1.410] |
| 2.41 | 13.3                 | 13.73                    | [0.978, 1.024] | [0.905, 1.134] |
| 5.00 | 7.2                  | 8.12                     | [0.975, 1.030] | [0.929, 1.108] |

**RESULT.** [Fisher and Monte Carlo agree] Where the quadratic approximation is valid the two estimates agree closely: 12.4 versus 12.37, 103.9 versus 98.75, 48.9 versus 45.64, 13.3 versus 13.73, 7.2 versus 8.12. They diverge only at  $B = 1$ , where both correctly report an unbounded result.



---

**CHECK.** [A withdrawn claim, recorded so it is not reintroduced] An earlier stage of this project reported a systematic factor-of-ten discrepancy between Fisher and Monte Carlo uncertainties. That was traced to finite-difference Jacobians contaminated by the solver noise floor and is **withdrawn**. With the analytic Jacobians of derivation 11.1 no such discrepancy exists, as the table above shows. The withdrawn figure must not appear as a scientific result anywhere in this project. Similarly, an earlier optimizer-initialisation study reported a factor-of-23 spread in recovered  $\tau_q$ ; it used the split-pulse inversion whose failure is characterised in section 21, with the failure time inside the sampling window. With the certified solver the effect is a factor of 4.7 rising to 9.3 at a one-log-unit perturbation [PREVIOUSLY VERIFIED]. The earlier figure is withdrawn.

---

## 16 Model selection by BIC

**Definition 16.1** (BIC convention used). For  $n$  observations,  $k$  free parameters and Gaussian errors of unknown but constant variance, the maximised log-likelihood satisfies  $-2 \ln \mathcal{L}_{\max} = n \ln(\text{SSE}/n) + \text{const}$ , giving the convention used throughout this work:

$$\boxed{\text{BIC} = n \ln\left(\frac{\text{SSE}}{n}\right) + k \ln n} \quad (119)$$

Only differences  $\Delta\text{BIC}$  between models fitted to the *same* data are interpreted, so the omitted additive constant is irrelevant.

---

**CHECK.** [Verification of the convention] With  $n = 80$ :  $\ln n = \ln 80 = 4.382$ . The penalty difference between GK ( $k = 3$ ) and Fourier ( $k = 1$ ) is  $(3 - 1) \times 4.382 = 8.764$ . Hence at exact resonance, where the two models fit identically well in expectation, BIC must prefer Fourier by approximately 8.8 units minus whatever the two extra parameters recover by fitting noise. The observed value 6.64 is consistent with this and is *not* an arbitrary number: it is the parsimony penalty partially offset by noise-fitting. [DERIVED VALUE]

---

**Calculation 16.1** (Protocol). Data generated by the full GK model at  $\eta = 2\%$ , configuration of definition 13.2. All four candidates of table 3 fitted to the same realisation with the same solver, same optimizer, same tolerances, and three restarts to reduce local-minimum risk. Averaged over 15 noise realisations.  $\Delta\text{BIC}$  is measured from the best model in each row. Script `validation/audit/run_bic.py`.

Table 8: BIC model selection. All values [NUMERICALLY COMPUTED HERE]. Lower is better; 0.00 marks the selected model.

| True $B$    | Fourier ( $k=1$ ) | MCV ( $k=2$ ) | Nyíri ( $k=2$ ) | GK ( $k=3$ ) | Selected       |
|-------------|-------------------|---------------|-----------------|--------------|----------------|
| <b>1.00</b> | <b>0.00</b>       | 236.58        | 4.03            | 6.64         | <b>Fourier</b> |
| 1.28        | 23.14             | 253.21        | 1.38            | <b>0.00</b>  | GK             |
| 2.41        | 166.73            | 302.12        | 44.21           | <b>0.00</b>  | GK             |
| 5.00        | 267.00            | 349.25        | 105.29          | <b>0.00</b>  | GK             |

---

**RESULT.** At  $B = 1$ , BIC selects Fourier over GK from data generated by GK, with  $\Delta\text{BIC}_{\text{GK}} = 6.64$ . This is the rank deficiency of proposition 9.1 recovered by a route that uses no derivative information and no Fisher matrix. Away from resonance GK is selected decisively. MCV is rejected everywhere by 237 to 349 units.

---

---

**CHECK.** [Interpretation limit — mandatory] BIC measures statistical parsimony under the stated data-generating process, noise model and observation design. It is used here as an identifiability diagnostic. It is *not* evidence that the material obeys Fourier’s law: at  $B = 1$  the two models are observationally identical, so the preference reports what the data can distinguish, not the underlying physics.

---

## 17 Published calibration calculations

Every value below is labelled. Values marked [LITERATURE VALUE] are transcribed from the cited primary source; values marked [DERIVED VALUE] are computed here from those literature values by (51). No value is transferred between sources without stating which source it came from.

### 17.1 Worked arithmetic for each case

**Calculation 17.1** ( $B$  from tabulated  $(\alpha, \tau_q, \kappa^2)$ ). Wherever a source tabulates all three of  $\alpha_{\text{GK}}$ ,  $\tau_q$  and  $\kappa^2$ ,  $B$  is obtained directly from (51). The units cancel: with  $\alpha$  in  $10^{-6} \text{ m}^2 \text{ s}^{-1}$ ,  $\tau_q$  in s and  $\kappa^2$  in  $10^{-6} \text{ m}^2$ , the factors  $10^{-6}$  cancel between numerator and denominator, so the tabulated numbers may be used directly. Explicitly, for the three Fehér–Kovács basalt specimens [8]:

$$B(1.86 \text{ mm}) = \frac{0.168}{0.61 \times 0.211} = \frac{0.168}{0.12871} = 1.3053, \quad (120)$$

$$B(2.75 \text{ mm}) = \frac{0.268}{0.61 \times 0.344} = \frac{0.268}{0.20984} = 1.2772, \quad (121)$$

$$B(3.84 \text{ mm}) = \frac{0.650}{0.68 \times 1.000} = \frac{0.650}{0.68} = 0.9559. \quad (122)$$

The same source reports 1.295, 1.272 and 0.94 respectively in its Table 2 [LITERATURE VALUE], so the largest disagreement is  $|0.9559 - 0.94| = 0.0159$ .

---

**CHECK.** Check 11b: the maximum  $|B_{\text{derived}} - B_{\text{reported}}|$  over all cases where both exist is 0.0159, occurring for the 3.84 mm specimen. Agreement to better than 0.02 in every case confirms that the present interpretation of the tabulated symbols matches the authors’ own. [NUMERICALLY COMPUTED HERE]

---

**Calculation 17.2** (Both et al. capacitor, from the primary source). Both et al. [2] report, for the layered capacitor of thickness 3.9 mm, from a nonlinear regression on 2250 data points:

$$\alpha = (1.958 \pm 0.004) \times 10^{-6} \text{ m}^2 \text{ s}^{-1} \quad [\text{LITERATURE VALUE}] \quad (123)$$

$$\tau_q = (0.51 \pm 0.01) \text{ s} \quad [\text{LITERATURE VALUE}] \quad (124)$$

$$\ell^2 = (1.53 \pm 0.03) \times 10^{-6} \text{ m}^2 \quad [\text{LITERATURE VALUE}] \quad (125)$$

$$R^2 = 0.9996 \quad [\text{LITERATURE VALUE}] \quad (126)$$

With  $\ell^2 \equiv \kappa^2$  (section 2.3),

$$B = \frac{1.53 \times 10^{-6}}{1.958 \times 10^{-6} \times 0.51} = \frac{1.53}{0.99858} = 1.5322. \quad [\text{DERIVED VALUE}] \quad (127)$$

Propagating the three independent relative errors in quadrature,

$$\begin{aligned} \frac{\delta B}{B} &= \sqrt{\left(\frac{0.03}{1.53}\right)^2 + \left(\frac{0.004}{1.958}\right)^2 + \left(\frac{0.01}{0.51}\right)^2} \\ &= \sqrt{(0.01961)^2 + (0.00204)^2 + (0.01961)^2} = 0.02780, \end{aligned} \quad (128)$$

so  $B = 1.532 \pm 0.043$  [DERIVED VALUE], a relative uncertainty of 2.78%.

---

**CHECK.** Checks 12c and 12d reproduce (127) and (128). [NUMERICALLY COMPUTED HERE]

---

**CHECK.** [Which dataset is which — mandatory distinction] Ván et al. [3] also report a capacitor of thickness 3.9 mm, with  $\alpha_{\text{GK}} = 2.13 \times 10^{-6}$ ,  $\tau = 0.954$  s and  $b = 2.23$  [LITERATURE VALUE]. These are *different fitted values* for the same specimen type, obtained by a different evaluation. They are listed as a separate case in table 9 and are never merged with (123)–(125). Only the Both et al. entry carries reported uncertainties.

---

## 17.2 The Ván et al. $\ell/b$ consistency question

**Calculation 17.3** (Reconciling the two tables of [3]). That source defines [3]

$$\hat{\tau} = \frac{\tau}{t_p}, \quad \hat{\alpha} = \frac{\alpha t_p}{L^2}, \quad \hat{\ell} = \frac{\ell}{L}, \quad \hat{b} = \frac{\hat{\ell}^2}{\hat{\tau} \hat{\alpha}} = b, \quad (129)$$

a different nondimensionalisation from definition 4.1 (time scaled by  $t_p$ , not  $L^2/\alpha$ ). Substituting,

$$b = \frac{(\ell/L)^2}{(\tau/t_p)(\alpha t_p/L^2)} = \frac{\ell^2}{L^2} \cdot \frac{t_p L^2}{\tau \alpha t_p} = \frac{\ell^2}{\tau \alpha}, \quad (130)$$

so  $b$  is algebraically identical to  $B$ , with both  $t_p$  and  $L$  cancelling. Evaluating  $\hat{\ell}^2/(\hat{\tau} \hat{\alpha})$  from that source’s *dimensionless* table alone gives 1.988, 3.031 and 1.782 for limestone, metal foam and leucocratic rock, against reported 2.17, 3.04 and 1.77 [DERIVED VALUE]: agreement to 0.01 for the latter two.

---

**CHECK.** [Root cause of the residual mismatch] Combining that source’s dimensionless table with its dimensional table does *not* reproduce  $b$ : for limestone the route  $\ell^2/(\tau \alpha_{\text{GK}})$  gives 0.219 against 2.17. Testing the internal consistency of the two tables through  $\hat{\alpha} \stackrel{?}{=} \alpha_{\text{GK}} t_p / L^2$  with  $t_p = 0.01$  s:

| Sample      | $\hat{\alpha}$ computed | $\hat{\alpha}$ reported | ratio |
|-------------|-------------------------|-------------------------|-------|
| Capacitor   | $1.400 \times 10^{-3}$  | $1.400 \times 10^{-3}$  | 1.000 |
| Limestone   | $1.021 \times 10^{-2}$  | $1.124 \times 10^{-3}$  | 9.08  |
| Metal foam  | $9.123 \times 10^{-4}$  | $9.120 \times 10^{-4}$  | 1.000 |
| Leucocratic | $1.558 \times 10^{-2}$  | $1.560 \times 10^{-3}$  | 9.98  |

Two rows agree to four significant figures; two differ by factors of 9.1 and 10.0 [DERIVED VALUE]. This is consistent with a units slip confined to two rows of the published table.

---

**RESULT.** The authors’ own reported  $b$  values are used in table 9 for cases 1–4, and the two affected rows are flagged. The discrepancy is documented, not silently recomputed, because resolving it definitively would require information not present in the published tables.

---

### 17.3 Master calibration table

Table 9: All twelve audited calibrations. R = reported by the source [LITERATURE VALUE]; C = calculated here [DERIVED VALUE] by (51). Band is (113), valid for the configuration of definition 13.2 only.  $\alpha$  in  $10^{-6} \text{ m}^2 \text{ s}^{-1}$ ,  $\tau_q$  in s,  $\kappa^2$  in  $10^{-6} \text{ m}^2$ .

| #  | Source             | Specimen            | $\alpha$   | $\tau_q$  | $\kappa^2$ | $B$               | In band? |
|----|--------------------|---------------------|------------|-----------|------------|-------------------|----------|
| 1  | Ván+ [3]           | capacitor 3.9 mm    | 2.13 R     | 0.954 R   | see §17.2  | 2.23 R            | no       |
| 2  | Ván+ [3]           | limestone 1.7 mm    | 2.950 R    | 0.991 R   | see §17.2  | 2.17 R            | no       |
| 3  | Ván+ [3]           | metal foam 5.1 mm   | 2.373 R    | 0.402 R   | see §17.2  | 3.04 R            | no       |
| 4  | Ván+ [3]           | leucocratic 1.75 mm | 4.77 R     | 1.32 R    | see §17.2  | 1.77 R            | yes      |
| 5  | earlier, in [8] T1 | basalt 1.86 mm      | 0.55 R     | 0.738 R   | 0.509 R    | 1.243 R / 1.254 C | yes      |
| 6  | earlier, in [8] T1 | basalt 2.75 mm      | 0.604 R    | 0.955 R   | 0.670 R    | 1.171 R / 1.162 C | yes      |
| 7  | earlier, in [8] T1 | basalt 3.84 mm      | 0.68 R     | 0.664 R   | 0.480 R    | 1.060 R / 1.063 C | yes      |
| 8  | Fehér-Kov. [8]     | basalt 1.86 mm      | 0.61 R     | 0.211 R   | 0.168 R    | 1.295 R / 1.305 C | yes      |
| 9  | Fehér-Kov. [8]     | basalt 2.75 mm      | 0.61 R     | 0.344 R   | 0.268 R    | 1.272 R / 1.277 C | yes      |
| 10 | Fehér-Kov. [8]     | basalt 3.84 mm      | 0.68 R     | 1.000 R   | 0.650 R    | 0.940 R / 0.956 C | yes      |
| 11 | Fehér-Kov. [8]     | foam 5.2 mm         | 3.01 R     | 0.304 R   | 2.203 R    | 2.395 R / 2.408 C | no       |
| 12 | Both et al. [2]    | capacitor 3.9 mm    | 1.958(4) R | 0.51(1) R | 1.53(3) R  | 1.532(43) C       | yes      |

**Aggregate:** 8 of 12 inside the band (113); 4 of 12 outside. **Reported uncertainty:** case 12 only; the other eleven report none. **Verification:** cases 5–7 verified against Table 1 of [8], which tabulates the earlier evaluation alongside the refined one, and Table 2 for their  $B$ ; they were not traced further back to the original measurement report. Cases 1–4 use the authors’ reported  $b$ ; see §17.2.

**CHECK.** Check 11a counts 8 of 12 inside [0.628, 1.804], matching the aggregate above. The eight are cases 4–10 and 12. [NUMERICALLY COMPUTED HERE]

### 17.4 Same-specimen discrepancy

Table 10: Two independent evaluations of the same three basalt specimens, both tabulated in [8]. All values [LITERATURE VALUE].

| Specimen       | $\tau_q$ earlier [s] | $\tau_q$ refined [s] | Ratio [DERIVED VALUE] | $R^2$ refined |
|----------------|----------------------|----------------------|-----------------------|---------------|
| basalt 1.86 mm | 0.738                | 0.211                | 3.50                  | 0.9975        |
| basalt 2.75 mm | 0.955                | 0.344                | 2.78                  | 0.9964        |
| basalt 3.84 mm | 0.664                | 1.000                | 1.51                  | 0.9986        |

**RESULT.** All six evaluations sit at  $B = 0.94$  to  $1.31$ . The profile-likelihood width at  $B = 1.28$  is a factor of 3.5 (table 6), which brackets the observed ratios 1.51 to 3.50. The spread is therefore consistent with and quantitatively accounted for by the identifiability structure near  $B \approx 1$ .

**No causal claim is made.** The two evaluations differ in pulse modelling, smoothing and fitting procedure as well as in conditioning, and the published information does not permit these contributions to be separated.

## 18 The Fehér–Kovács fitting domain

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**RESULT.** [Verbatim from the primary source] Fehér and Kovács state [8]: “we still had to restrict ourselves to a domain, which is  $3 > \kappa^2/(\alpha\tau_q) \geq 1$ . Its lower limit expresses the Fourier case.” In the notation of this document,

$$1 \leq B < 3. \quad (131)$$

[LITERATURE VALUE]

---

**Significance of the lower boundary.** By proposition 8.1,  $B = 1$  is exactly the condition at which the GK response becomes the Fourier response, and by proposition 9.1 it is exactly where  $(\tau_q, \kappa^2)$  become structurally non-identifiable. The accepted evaluation domain therefore has the degeneracy as its lower boundary.

---

**CHECK.** [What is *not* being asserted] The restriction (131) is well motivated: room-temperature measurements on these heterogeneous specimens indicate over-diffusive behaviour, so the domain reflects the observed regime, and bounding a nonlinear search improves its stability. **It is not claimed that the procedure of [8] is wrong.** Nor is it claimed that reported values “happen to” fall near  $B = 1$ : the proximity is in part a consequence of the admissible domain, and must be described as such. The same source explicitly notes that GK theory itself is not restricted to this domain.

---

## 19 Calibration uncertainty consistency check

Case 12 is the only audited calibration reporting parameter uncertainties, so it is the only opportunity to test the Fisher machinery of section 11 against a real published dataset.

**Calculation 19.1** (Published relative uncertainty). From (124):

$$\frac{\delta\tau_q}{\tau_q} = \frac{0.01}{0.51} = 0.019608 = 1.96\%. \quad \text{PUBLISHED [LITERATURE VALUE](arithmetic [DERIVED VALUE])} \quad (132)$$

**Calculation 19.2** (Predicted uncertainty under their design). **Assumptions, stated explicitly.** The Fisher calculation of calculation 13.1 is repeated with:

$$B = 1.532 \quad \text{from (127),} \quad (133)$$

$$n = 2250 \quad \text{data points, as stated in [2][LITERATURE VALUE],} \quad (134)$$

$$\eta = 1\% \quad \text{assumed noise level (not stated in the source),} \quad (135)$$

$$\text{all other settings as in definition 13.2.} \quad (136)$$

---

**RESULT.**

$$\text{s. e.}(\tau_q)|_{\text{their design}} = 2.55\%. \quad \text{CALCULATED HERE [NUMERICALLY COMPUTED HERE]} \quad (137)$$

Check 12a, script `validation/calc/check_numerical.py`. Compare with the published 1.96% of (132).

---

---

**CHECK.** [Interpretation — deliberately limited] The two figures agree to within about 30% of each other, from wholly independent routes: one an asymptotic regression error reported by the original authors, the other a Fisher prediction from the framework of this document.

**This is an external consistency check against a published real dataset, not a validation.**

The noise level (135) is assumed rather than taken from the source, the sampling times are assumed uniform, and the reported errors are themselves asymptotic and computed, as the authors state, assuming exact  $L$  and  $t_p$ . The check should be read as confirming that the framework produces uncertainties of the right order for a real design, and as demonstrating concretely why the band (113) is configuration-dependent: the same  $B$  with 2250 low-noise points behaves very differently from 80 points at 2% noise.

---

## 20 Validation anchor: reproduction of published results

**Definition 20.1** (Anchor). R. Kovács, *Analytic solution of Guyer–Krumhansl equation for laser flash experiments*, International Journal of Heat and Mass Transfer **127**, Part A, 631–636 (2018), DOI 10.1016/j.ijheatmasstransfer.2018.06.082 [1]. The version of record was confirmed on the publisher page; sections 4–5 are paywalled and the equations used here were read from the open preprint arXiv:1804.05225, whose equation numbering differs from the version of record by one.

**Calculation 20.1** (Reproduction procedure). (1) Published figures extracted at native resolution ( $4000 \times 1908$  px).

- (2) Two independent digitisation passes: intensity-weighted centroid of the curve band, and envelope midpoint.
- (3) Digitisation noise floor

$$\sigma_d = \sqrt{(\Delta_{AB})^2 + \text{median}(\text{half-width})^2}, \quad (138)$$

where  $\Delta_{AB}$  is the pointwise difference between the passes.

- (4) Published parameters used verbatim. **No fitting, no tuning, no adjustment of any kind.**
- (5) Solver: the certified convolution solver of section 21.
- (6) Metric:  $\text{NRMSE} = 100 \times \text{RMSE} / (\max T_{\text{pub}} - \min T_{\text{pub}})$ .

### 20.1 Truncated eigenfunction series required by Fig. 2

**Defect closed during Phase 3.** The convolution solver of section 21 returns the *converged* solution. The anchor’s Fig. 2 is a convergence study showing the solution *truncated* at  $N = 1, 3, 10, 40$  eigenfunction modes, so reproducing it requires the truncated series, which was not previously stated in this document. It is derived here.

**Derivation 20.1** (Truncated series at the Fourier-resonance parameters). Fig. 2 uses  $\tau_q = \kappa^2 = 0.02$ , i.e.  $B = 1$ , so by (66) the operator is exactly the Fourier operator. Expand the dimensionless temperature in the cosine basis  $X_n(x) = \cos(n\pi x)$ , which satisfies the insulated rear-face condition and is the eigenbasis of  $\partial_{xx}$  with eigenvalues  $-\beta_n$ ,  $\beta_n = (n\pi)^2$ :

$$T(x, t) = a_0(t) + \sum_{n \geq 1} a_n(t) \cos(n\pi x). \quad (139)$$

Projecting (28) on mode 0 gives  $\tau_\Delta \dot{a}_0 = q_0(t)$ , hence with  $t_c = \min(t, \tau_\Delta)$

$$a_0(t) = \frac{1}{\tau_\Delta} \int_0^{t_c} q_0(u) \, du = \frac{1}{\tau_\Delta} \left[ t_c - \frac{\sin(\omega t_c)}{\omega} \right], \quad \omega = \frac{2\pi}{\tau_\Delta}, \quad (140)$$

which equals 1 for  $t \geq \tau_\Delta$ , consistent with (23). Mode  $n \geq 1$  obeys  $\dot{a}_n + \beta_n a_n = A_n q_0(t)$  with  $A_n = 2(-1)^n/\tau_\Delta$  at the rear face  $x = 1$ , where  $\cos(n\pi) = (-1)^n$ . Solving by the integrating factor and evaluating the pulse integral only over its support gives

$$a_n(t) = A_n [I_1(t) - I_2(t)], \quad (141)$$

$$I_1 = \frac{e^{-\beta_n(t-t_c)} - e^{-\beta_n t}}{\beta_n}, \quad I_2 = \frac{e^{-\beta_n(t-t_c)} [\beta_n \cos(\omega t_c) + \omega \sin(\omega t_c)] - \beta_n e^{-\beta_n t}}{\beta_n^2 + \omega^2}. \quad (142)$$

The rear-face solution truncated at  $N$  modes is therefore

$$T_N(1, t) = a_0(t) + \sum_{n=1}^N a_n(t) \quad (143)$$

---

**CHECK.** Script `validation/calc/derive_series.py` verifies symbolically that the closed form  $I_1 - I_2$  of (142) equals the direct integral  $e^{-\beta_n t} \int_0^{t_c} e^{\beta_n u} q_0(u) du$ : residual exactly 0. It also confirms  $a_0(\tau_\Delta) = 1$ . [\[NUMERICALLY COMPUTED HERE\]](#)

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**CHECK.** [Scope] (143) is used *only* to reproduce the anchor's truncation study (test T4). Every other result in this document uses the converged convolution solver of section 21. The two must agree as  $N \rightarrow \infty$ , which is test T4b.

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Table 11: Reproduction parameters, taken verbatim from [1] [\[LITERATURE VALUE\]](#).  $\tau_\Delta = 0.04$  in every case.

| Anchor figure | Regime            | $\tau_q$ | $\kappa^2$ | Series terms / span                 |
|---------------|-------------------|----------|------------|-------------------------------------|
| Fig. 2        | convergence       | 0.02     | 0.02       | $N = 1, 3, 10, 40$ ; $t \in [0, 1]$ |
| Fig. 3        | $B = 1$ resonance | 0.02     | 0.02       | $N = 40$ ; $t \in [0, 1]$           |
| Fig. 5        | over-diffusive    | 0.02     | 0.20       | $N = 10$ ; $t \in [0, 2]$           |

---

**RESULT.** [Reproduction errors]

| Figure | Points | RMSE    | NRMSE         | $\sigma_d$ | RMSE/ $\sigma_d$ | within $\pm 2\sigma_d$ |
|--------|--------|---------|---------------|------------|------------------|------------------------|
| Fig. 3 | 729    | 0.00384 | <b>0.384%</b> | 0.00593    | 0.65             | 99.9%                  |
| Fig. 5 | 743    | 0.00288 | <b>0.382%</b> | 0.00603    | 0.48             | 99.5%                  |
| Fig. 2 | 1999   | 0.00819 | <b>0.483%</b> | —          | —                | —                      |

All [\[NUMERICALLY COMPUTED HERE\]](#); checks 13-fig3 and 13-fig5 reproduce the first two rows exactly. Fig. 2 is the pooled value over four truncation levels [\[PREVIOUSLY VERIFIED\]](#).

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**CHECK.** [Reading the ratio  $\text{RMSE}/\sigma_d < 1$ ] The reproduction error is *below* the digitisation noise floor: the residual is dominated by the precision with which the published curves can be read, not by any discrepancy between the published solution and the present implementation. A closer agreement could not be demonstrated from published figures.

**Normalisation artefact, reported openly.** The  $N = 40$  curve of Fig. 2 is nearly converged and spans a range of only 0.031, so normalising its residual by its own range gives 8.19%, while the same residual normalised by the full range of the figure is 0.162%; the absolute error,  $2.7 \times 10^{-3}$ , is comparable to the other curves [\[PREVIOUSLY VERIFIED\]](#).

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**RESULT.** [Scope of this validation — mandatory] This exercise validates the *forward computational implementation* against an independently authored published solution. It does **not** validate the inverse problem, and no such claim is made. The inverse results of sections 9 and 13 rest on the analytic proof, the arbitrary-precision test of section 10, and the independent statistical route of section 16.

Distinguish carefully:

- **External validation:** agreement with results published by another author (this section).
  - **Numerical cross-checking:** agreement between independent implementations of our own equations (section 21). This is not external validation.
- 

## 21 Numerical certification

### 21.1 Factorisation and why the pulse is not split

**Derivation 21.1** (Kernel/pulse factorisation). The rear-face transform (47) factorises as

$$\hat{T}(1, s) = \hat{K}(s) \hat{Q}_0(s), \quad \hat{K}(s) = \frac{m(s)}{\tau_\Delta s \sinh m(s)}, \quad (144)$$

a material kernel times the pulse transform.  $\hat{K}$  has no poles on the imaginary axis;  $\hat{Q}_0$  has the removable ones of section 5. Therefore the kernel alone is inverted numerically and the temperature is recovered by exact convolution:

$$T(1, t) = \int_0^{\min(t, \tau_\Delta)} k(t-u) q_0(u) du, \quad q_0(u) = 1 - \cos\left(\frac{2\pi u}{\tau_\Delta}\right), \quad (145)$$

$k$  being the inverse transform of  $\hat{K}$ .

**Derivation 21.2** (The failure mode of the split formulation). Suppose instead the pulse factor  $1 - e^{-s\tau_\Delta}$  is handled by the second-shift theorem and  $\omega^2/[s(s^2 + \omega^2)]$  is inverted alone. That factor retains genuine poles at  $s = \pm i\omega$ . The fixed Talbot rule evaluates the transform at

$$s_j = r \theta_j (\cot \theta_j + i), \quad \theta_j = \frac{j\pi}{M}, \quad r = \frac{2M}{5t}. \quad (146)$$

$\text{Re } s_j = r \theta_j \cot \theta_j = 0$  requires  $\cot \theta_j = 0$ , i.e.  $\theta_j = \pi/2$ , i.e.  $j = M/2$ , which is an integer *only when  $M$  is even*. At that node  $s = i r \pi/2$ . This coincides with the pole  $s = i\omega$  when

$$\frac{r\pi}{2} = \omega = \frac{2\pi}{\tau_\Delta} \implies \frac{2M}{5t} \cdot \frac{\pi}{2} = \frac{2\pi}{\tau_\Delta} \implies \boxed{t^* = \frac{M \tau_\Delta}{10}} \quad (147)$$

---

**CHECK.** [Prediction tested directly] Check 7c verifies (147) symbolically. Numerically, with  $\tau_\Delta = 0.04$  [PREVIOUSLY VERIFIED]:

| $M$       | $t^*$  | error at $0.98t^*$   | error at $t^*$        | error at $1.02t^*$   |
|-----------|--------|----------------------|-----------------------|----------------------|
| 20 (even) | 0.0800 | $1.9 \times 10^{-4}$ | $1.9 \times 10^{+11}$ | $1.0 \times 10^{-3}$ |
| 40 (even) | 0.1600 | $3.9 \times 10^{-5}$ | $9.7 \times 10^{+10}$ | $7.2 \times 10^{-4}$ |
| 41 (odd)  | 0.1640 | $1.1 \times 10^{-4}$ | $2.6 \times 10^{-4}$  | $2.1 \times 10^{-4}$ |
| 60 (even) | 0.2400 | $6.8 \times 10^{-5}$ | $5.3 \times 10^{+9}$  | $5.7 \times 10^{-4}$ |
| 64 (even) | 0.2560 | $9.2 \times 10^{-5}$ | $8.4 \times 10^{+10}$ | $5.8 \times 10^{-4}$ |

The error rises by ten to eleven orders of magnitude at exactly the predicted time for every even  $M$ , and not at all for odd  $M$ .

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**RESULT.** [Corrected certification — supersedes earlier statements]

- (1) The failure is a property of the numerical inversion, *not* of the physics: the exact solution is regular at  $t^*$ .
  - (2) The remedy adopted is the convolution formulation (145), which inverts only the pole-free kernel, so the removable singularities never appear.
  - (3) Additionally  $M$  is chosen **odd**, which removes the quadrature node from the imaginary axis entirely. Production value  $M = 41$ .
  - (4) The naive remedy of increasing  $M$  is actively harmful: by (147) the failure time scales with  $M$  and simply moves, while round-off grows.
  - (5) **An earlier certification described a “safe window”  $M \in [32, 48]$  with instability at  $M \approx 56$ – $68$ . That description was obtained on a six-point probe set and does not survive testing on a dense grid; it is superseded by items (1)–(4) and must not be restored.**
  - (6) With the convolution formulation the accuracy plateau extends across  $M \in [25, 81]$ , degrading beyond  $M \approx 91$  through round-off [PREVIOUSLY VERIFIED].
- 

## 21.2 Convergence, precision and cross-solver checks

Table 12: Numerical certification and error budget. Cross-solver agreement is numerical cross-checking, not external validation.

| Item                                           | Result                                                          | Scope / label                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Convolution quadrature                         | $\leq 10^{-13}$                                                 | 32 versus 64 Gauss–Legendre nodes [PREVIOUSLY VERIFIED]                                                     |
| Solver A vs. 50-digit reference                | $10^{-11}$ to $10^{-13}$                                        | probe times away from the steep rise [PREVIOUSLY VERIFIED]                                                  |
| Solver A vs. independent FD                    | $\approx 10^{-7}$                                               | full record; limited by the FD discretisation [PREVIOUSLY VERIFIED]                                         |
| FD spatial convergence                         | observed order $p = 2.00$ ; $2.6 \times 10^{-6}$ at $N_x = 400$ | Richardson reference [PREVIOUSLY VERIFIED]                                                                  |
| FD temporal                                    | $\approx 10^{-9}$                                               | relative tolerance $10^{-10}$ [PREVIOUSLY VERIFIED]                                                         |
| Arithmetic precision                           | $\leq 8.9 \times 10^{-16}$                                      | 15, 30 and 50 digits compared [PREVIOUSLY VERIFIED]                                                         |
| Fourier-resonance limit                        | $2.6 \times 10^{-6}$                                            | $\kappa^2 = \alpha\tau_q$ against the analytical Fourier solution [PREVIOUSLY VERIFIED]                     |
| Fourier limit $\tau_q, \kappa^2 \rightarrow 0$ | $2.6 \times 10^{-6}$                                            | against the analytical Fourier solution [PREVIOUSLY VERIFIED]                                               |
| Non-negativity                                 | $T \geq 0$ in all cases tested                                  | contrasts with negative-temperature artefacts reported for dual-phase-lag models [12] [PREVIOUSLY VERIFIED] |
| Steady state                                   | $T(1, t \rightarrow \infty) = 1.00000000$                       | confirms the normalisation (23) [PREVIOUSLY VERIFIED]                                                       |
| <b>Total numerical error</b>                   | $\approx 10^{-6}$                                               | versus digitisation floor $6 \times 10^{-3}$                                                                |

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**RESULT.** The total numerical error of the framework is about  $10^{-6}$ , three orders of magnitude below the digitisation noise floor of section 20. The reproduction reported there is therefore digitisation-limited, not solver-limited.

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## 22 Master traceability table

Table 13: Traceability of every principal result. “Program check” marks the items the Phase-3 program must reproduce and the Phase-5 comparison must confirm.

| Item                             | Eq./Tab.       | Origin                                      | Derived here? | Num. checked?      | Program check |
|----------------------------------|----------------|---------------------------------------------|---------------|--------------------|---------------|
| Energy balance                   | (10)           | literature [1]                              | no            | n/a                | no            |
| GK constitutive law              | (11)           | literature [4, 5, 6]                        | no            | n/a                | no            |
| Flux equation                    | (14)           | derived                                     | yes           | n/a                | no            |
| Pulse BC                         | (15)           | anchor [1]                                  | no            | n/a                | yes           |
| Derived IC                       | (20)           | derived                                     | yes           | n/a                | yes           |
| $\partial_t q(x, 0) = 0$         |                |                                             |               |                    |               |
| Scaling                          | (21)           | anchor [1]                                  | reproduced    | n/a                | yes           |
| $\bar{q}_0 = q_{\max}$           | (22)           | derived                                     | yes           | check 7d           | yes           |
| $T_{\text{end}}$                 | (23)           | anchor [1]                                  | reproduced    | n/a                | yes           |
| Dimensionless sys-tem            | (28),(33)      | derived                                     | yes           | n/a                | yes           |
| $\hat{\alpha} \equiv 1$          | derivation 4.4 | derived                                     | yes           | n/a                | yes           |
| Laplace elimination              | (41)–(43)      | derived                                     | yes           | check 1a           | no            |
| <b>Propagation factor</b>        | (44)           | derived                                     | yes           | checks 1a,1b,2a–2c | yes           |
| Spatial solution                 | (45)           | derived                                     | yes           | checks 5c,5d       | yes           |
| Rear-face transform              | (47)           | derived                                     | yes           | check 5d           | yes           |
| Pulse transform                  | (48)           | derived                                     | yes           | check 7a           | yes           |
| Removable poles                  | (49)           | derived                                     | yes           | check 7b           | yes           |
| $B$ definition                   | (51)           | adopted; ratio used by [8, 3]               | no            | check 4a           | yes           |
| $B$ scale invariance             | (53)           | derived                                     | yes           | check 4d           | yes           |
| $\kappa^2 = \alpha\tau_q B$      | (55)           | derived                                     | yes           | check 4b           | yes           |
| Param. Jacobian = $\alpha\tau_q$ | (57)           | derived                                     | yes           | check 4c           | no            |
| Fourier limit                    | (58)           | derived                                     | yes           | check 3a           | yes           |
| MCV limit                        | (59)           | derived                                     | yes           | check 3b           | yes           |
| MCV wave speed                   | (60)           | derived                                     | yes           | check 3d           | no            |
| Nyíri limit                      | (61)           | derived; [7]                                | yes           | check 3c           | yes           |
| <b>Resonance cancellation</b>    | (66)           | derived; condition is prior art [10, 1, 11] | yes           | checks 5a,5b       | yes           |
| $\tau_q$ -independence on ray    | (69)           | derived                                     | yes           | checks 5c,5d       | yes           |
| $\partial m / \partial \tau_q$   | (71)           | derived                                     | yes           | check 5e           | yes           |
| $\partial m / \partial \kappa^2$ | (72)           | derived                                     | yes           | check 5f           | yes           |
| General ratio $-s/m^2$           | (74)           | derived                                     | yes           | check 5g           | no            |
| <b>Ratio</b> = $-\alpha$         | (75)           | derived                                     | yes           | checks 5h,6a       | yes           |
| Anti-parallel sensitivities      | (76)           | derived                                     | yes           | check 6a           | yes           |

| Item                          | Eq./Tab.          | Origin                          | Derived here? | Num. checked?         | Program check |
|-------------------------------|-------------------|---------------------------------|---------------|-----------------------|---------------|
| Rank one, null $(1, \alpha)$  | (80),(81)         | derived                         | yes           | check 6b              | yes           |
| $\det F = 0$                  | (85)              | derived                         | yes           | check 6c              | yes           |
| $\text{corr} = -1$            | (86)              | derived                         | yes           | check 6d              | yes           |
| $\partial_m \hat{K}$          | (98)              | derived                         | yes           | verified              | yes           |
| Reparam. degeneracy           | (102)             | derived                         | yes           | check 6b              | yes           |
| Numerical degeneracy          | (89)–(91)         | computed                        | n/a           | checks 7a–7c          | yes           |
| 50-digit degeneracy           | (92)              | computed                        | n/a           | check 7d              | yes           |
| Fisher landscape              | table 5           | computed                        | n/a           | checks 8-*            | yes           |
| <b>Band</b> $[0.628, 1.804]$  | (113)             | computed                        | n/a           | checks 8f,8g          | yes           |
| Other bands                   | (112),(114),(115) | computed                        | n/a           | same method           | yes           |
| Profile likelihood            | table 6           | computed                        | n/a           | rerun                 | yes           |
| Monte Carlo                   | table 7           | computed                        | n/a           | rerun                 | yes           |
| BIC convention                | (119)             | standard                        | no            | penalty check         | yes           |
| BIC results                   | table 8           | computed                        | n/a           | rerun                 | yes           |
| $B$ per calibration           | table 9           | derived from [LITERATURE VALUE] | yes           | check 11a,11b         | no            |
| Both et al. $B$               | (127)             | derived from [LITERATURE VALUE] | yes           | check 12c             | no            |
| Both et al. $\delta B$        | (128)             | derived                         | yes           | check 12d             | no            |
| Uncertainty cross-check       | (137)             | computed                        | n/a           | check 12a             | yes           |
| Fitting domain $1 \leq B < 3$ | (131)             | literature [8]                  | no            | n/a                   | no            |
| Anchor NRMSE                  | section 20        | computed                        | n/a           | checks 13-*           | yes           |
| $t^* = M\tau_\Delta/10$       | (147)             | derived                         | yes           | check 7c              | yes           |
| Error budget                  | table 12          | computed                        | n/a           | [PREVIOUSLY VERIFIED] | yes           |

## 23 Program implementation specification

This section specifies the Phase-3 program. **The program must implement exactly the equations of this document and must contain no physics or equation that does not appear above.**

### 23.1 Module 1: forward solver

#### Inputs

$\alpha$  [ $\text{m}^2 \text{s}^{-1}$ ],  $\tau_q$  [s],  $\kappa^2$  [ $\text{m}^2$ ],  $L$  [m],  $t_p$  [s], observation times  $t_i$  [s].

#### Internal scaling

$\hat{t}_i = \alpha t_i / L^2$ ,  $\hat{\tau}_q = \alpha \tau_q / L^2$ ,  $\hat{\kappa}^2 = \kappa^2 / L^2$ ,  $\hat{\tau}_\Delta = \alpha t_p / L^2$ , and  $\hat{\alpha} \equiv 1$  (derivation 4.4). **The**

**program must never mix  $\hat{\alpha} \equiv 1$  with the physical  $\alpha$  in one expression;** this specific error produced a  $10^6$  mis-scaling in earlier work.

**Kernel**

$\hat{K}(s) = m/(\hat{\tau}_\Delta s \sinh m)$  with  $m = \sqrt{s(1 + \hat{\tau}_q s)/(1 + \hat{\kappa}^2 s)}$ , principal branch (definition 5.2).

**Inversion**

fixed Talbot, nodes (146),  $M$  **odd, default**  $M = 41$ . The program must reject even  $M$  or emit a warning citing (147).

**Pulse handling**

convolution (145) with Gauss–Legendre, default 32 nodes. **The second-shift/split formulation is prohibited** (section 21.1).

**Overflow guards**

evaluate  $1/\sinh m$  as  $2e^{-m}/(1 - e^{-2m})$  and  $\coth m$  as  $(1 + e^{-2m})/(1 - e^{-2m})$ ; guard the removable  $0/0$  when  $e^{-2m}$  underflows.

**Output**

$T(1, t_i)$ , dimensionless, normalised so  $T \rightarrow 1$ .

## 23.2 Module 2: analytic Jacobian

**Outputs**

$\partial T/\partial \alpha$ ,  $\partial T/\partial \tau_q$ ,  $\partial T/\partial \kappa^2$  at each  $t_i$ .

**Method**

$\tau_q$  and  $\kappa^2$  analytically via (98) and (99), inverted and convolved by the same route as the forward solution.  $\alpha$  by a single central difference, since it rescales the time axis and pulse length rather than entering the kernel.

**Prohibited**

finite differencing of the time-domain solver for  $\tau_q$  or  $\kappa^2$  (section 15).

**Acceptance**

maximum relative deviation from Richardson-extrapolated central differences  $< 10^{-6}$ .

## 23.3 Module 3: independent cross-check solver

Staggered-grid method of lines:  $T$  at  $N_x$  cell centres,  $q$  at  $N_x + 1$  faces, so (34) sits exactly on grid points; BDF integrator,  $N_x = 400$ ,  $\text{rtol } 10^{-10}$ ,  $\text{atol } 10^{-12}$ . Must share no code with Module 1. Its agreement with Module 1 is numerical cross-checking, not validation.

## 23.4 Module 4: estimation and statistics

**Estimator**

Levenberg–Marquardt on  $\ln \theta$  with the Module-2 Jacobian;  $\text{xtol} = \text{ftol} = 10^{-13}$ .

**Fisher**

$\tilde{F} = \tilde{J}^\top \tilde{J}/\sigma^2$ , (95),(83); report  $\sqrt{\text{diag } \tilde{F}^{-1}}$ ,  $\text{cond } \tilde{F}$ , correlation, and  $\text{s.e.}(B)$  by the delta method (103).

**Profile likelihood**

(116)–(118);  $\geq 53$  log-spaced nodes over at least  $[10^{-1.3}, 10^{1.3}] \times \tau_q$ ,  $\geq 2$  restarts per node.

**Monte Carlo**

$\geq 200$  trials, fixed seeds, robust spread (IQR/1.349) reported alongside percentiles.

**BIC**

(119) with  $k$  from table 3; identical protocol across all four models.

## 23.5 Required validation tests and expected values

Table 14: Acceptance tests the Phase-3 program must pass. Phase-5 compares program output against this column.

| #   | Test                                                             | Expected                     | Tolerance          |
|-----|------------------------------------------------------------------|------------------------------|--------------------|
| T1  | $m^2$ at $\kappa^2 = \alpha\tau_q$ equals $s/\alpha$             | exact                        | $10^{-14}$         |
| T2  | Anchor Fig. 3 NRMSE                                              | 0.384%                       | $\pm 0.005$        |
| T3  | Anchor Fig. 5 NRMSE                                              | 0.382%                       | $\pm 0.005$        |
| T4  | Anchor Fig. 2 pooled NRMSE, series (143)                         | 0.483%                       | $\pm 0.01$         |
| T4b | Series (143) at large $N$ vs convolution solver                  | agree                        | $10^{-4}$          |
| T5  | Spread on ray, $\tau_q$ over $100\times$ , double precision      | $\leq 10^{-9}$               | —                  |
| T6  | Control spread at $B = 1.05$                                     | $\approx 2.6 \times 10^{-2}$ | $\pm 20\%$         |
| T7  | Fisher s.e. ( $\tau_q$ ) at $B = 0.5$                            | 12.37%                       | $\pm 0.1$          |
| T8  | Fisher s.e. ( $\tau_q$ ) at $B = 1.28$                           | 45.64%                       | $\pm 0.5$          |
| T9  | Band edges at 20%                                                | 0.628, 1.804                 | $\pm 0.005$        |
| T10 | cond $\tilde{F}$ at $B = 1 + 10^{-7}$                            | $> 10^{12}$                  | —                  |
| T11 | BIC at $B = 1$ : Fourier 0.00, GK                                | 6.64                         | $\pm 1.5$          |
| T12 | BIC at $B = 2.41$ selects GK                                     | GK                           | —                  |
| T13 | Profile factor at $B = 1.28$                                     | 3.5                          | $\pm 0.5$          |
| T14 | Profile at $B = 1$ exceeds grid                                  | unbounded                    | —                  |
| T15 | Monte Carlo vs Fisher at $B = 0.5$                               | agree within 20%             | —                  |
| T16 | Even- $M$ split solver fails at $t^* = M\tau_\Delta/10$          | error $> 10^6$               | —                  |
| T17 | Odd- $M$ convolution solver at same $t$                          | error $< 10^{-3}$            | —                  |
| T18 | FD vs Laplace solver, full record                                | $\leq 10^{-6}$               | —                  |
| T19 | Fourier limit $\tau_q, \kappa^2 \rightarrow 0$                   | $2.6 \times 10^{-6}$         | order of magnitude |
| T20 | $T(1, t \rightarrow \infty)$                                     | 1.00000000                   | $10^{-7}$          |
| T21 | $T \geq 0$ everywhere                                            | true                         | —                  |
| T22 | $B$ identical across $L = 1.86, 2.75, 3.84$ mm at fixed material | identical                    | $10^{-12}$         |
| T23 | $B$ for Both et al. values                                       | 1.532                        | $\pm 0.001$        |
| T24 | s.e. ( $\tau_q$ ) at $B = 1.532$ , $n = 2250$ , $\eta = 1\%$     | 2.55%                        | $\pm 0.2$          |

## 23.6 Required figures

F1 degeneracy on the ray with the  $B = 1.05$  control; F2 anchor reproduction overlays with residuals against  $\pm 2\sigma_d$ , labelled as reproduction; F3 Talbot pole-collision demonstration, even versus odd  $M$ ; F4 Fisher standard errors and cond  $\tilde{F}$  versus  $B$  with the band shaded; F5 profile likelihood curves with the  $\chi^2_{1,0.95}$  threshold; F6 BIC bar chart over the four models and four  $B$  values; F7 the twelve calibrations in  $B$ -space with the band and the domain (131) shaded.

## 23.7 Output format

Machine-readable JSON for every table in this document, keyed by the equation or table label, plus PDF figures. Phase-5 compares the JSON against table 14 automatically.

## 24 Final calculation audit

Table 15: Audit register. Full records in `validation/calc/checks_symbolic.json` and `checks_numerical.json`.

| #  | Check                      | Input                                          | Expected / Actual                                                        | Status |
|----|----------------------------|------------------------------------------------|--------------------------------------------------------------------------|--------|
| 1  | Symbolic algebra, $m^2$    | field equations                                | $s(1 + \tau_q s)/(1 + \kappa^2 s)$ / identical                           | PASS   |
| 2  | Dimensional analysis       | all terms of (10),(11),(44)                    | consistent / consistent                                                  | PASS   |
| 3  | Limiting cases             | $\tau_q \rightarrow 0, \kappa^2 \rightarrow 0$ | Fourier, MCV, Nyíri / all three recovered                                | PASS   |
| 4  | Parameter transformation   | $\theta \leftrightarrow \phi$                  | $\det = \alpha\tau_q / \alpha\tau_q$                                     | PASS   |
| 5  | Structural identifiability | $\kappa^2 = \alpha\tau_q$                      | $\partial_{\tau_q} T = 0$ , ratio = $-\alpha$ / both exact               | PASS   |
| 6  | Fisher rank                | $J$ on the ray                                 | rank 1, $\det F = 0$ , corr $-1$ / all three                             | PASS   |
| 7  | Numerical degeneracy       | $\tau_q$ over $100\times$                      | $\leq 10^{-9}$ / $1.26 \times 10^{-11}$ ; 50-digit $7.7 \times 10^{-30}$ | PASS   |
| 8  | Profile likelihood         | 53-node grid                                   | monotone widening; unbounded at $B = 1$ / as expected                    | PASS   |
| 9  | Monte Carlo                | 200 trials                                     | agrees with Fisher / 12.4 vs 12.37 etc.                                  | PASS   |
| 10 | BIC                        | 4 models, 15 realisations                      | Fourier at $B = 1$ / 0.00 vs GK 6.64                                     | PASS   |
| 11 | Calibration arithmetic     | 12 cases                                       | 8 inside; max dev 0.016 / 8; 0.0159                                      | PASS   |
| 12 | Both et al. uncertainty    | their design                                   | $\approx 2.5\%$ vs published 1.96% / 2.55%                               | PASS   |
| 13 | Anchor reproduction        | published params                               | 0.384%, 0.382% / identical                                               | PASS   |
| 14 | Stale-number scan          | superseded values                              | absent / absent                                                          | PASS   |
| 15 | Citation-source check      | 14 entries                                     | all resolve / all resolve                                                | PASS   |
| 16 | Equation-source check      | all numbered equations                         | origin assigned / assigned                                               | PASS   |

### 24.1 Discrepancies found and resolved during this audit

- (D1) **Sympy radical reduction.** Checks 6a, 6b and 6d initially failed because the algebra system does not reduce  $\sqrt{s^2\tau_q^2 + 2s\tau_q + 1}$  to  $(1 + s\tau_q)$ . The polynomial factors exactly as  $(1 + s\tau_q)^2$  and both factors are positive for  $s, \tau_q > 0$ , so the reduction is valid; numerical spot checks confirm the difference is zero to machine precision. The check script now applies the reduction explicitly and all three pass. **This was a limitation of the tool, not an error in the mathematics.**
- (D2) **Tuple comparison in the audit recorder.** The dimensional checks initially raised a type error because dimension pairs were compared with symbolic subtraction. Fixed in the recorder; no mathematical content affected.
- (D3) **Superseded Talbot certification.** An earlier “safe window  $M \in [32, 48]$ , unstable  $M \approx 56$ – $68$ ” was obtained on a six-point probe set and fails on a dense grid. Replaced by the derived pole-collision mechanism (147) and the convolution remedy (section 21.1).
- (D4) **Withdrawn Fisher/Monte-Carlo discrepancy and optimizer spread.** Both traced to numerical artefacts and withdrawn; recorded in section 15 so they are not reintroduced.

### 24.2 Stale-number scan

The following superseded values were searched for and are **absent** from this document except where explicitly identified as withdrawn: 0.661, 1.723 (old band); “6 of 11”, “7 of 11” (old aggregates); factor 23 (old optimizer spread); tenfold Fisher/Monte-Carlo discrepancy;  $B \leq 0.20$  (old design branch);  $M \in [32, 48]$  as a safe window.

### 24.3 Remaining blockers

- (B1) **Authenticated citation-index search not performed.** Scopus and Web of Science are authentication-walled in the working environment. This affects only statements about the prior literature, and this document makes none: it records calculations, not novelty claims. It remains a blocker for the Phase-6 manuscript.

- (B2) **Anchor sections 4–5 paywalled.** Equations were read from the open preprint [arXiv:1804.05225](https://arxiv.org/abs/1804.05225); the bibliographic record was confirmed against the publisher page. Equation numbers of the anchor are therefore never cited.
- (B3) **Raw heat-pulse time series unavailable.** The calibration audit of section 17 necessarily works from tabulated fitted values. A direct re-inversion of the original records is not possible with published material.
- (B4) **Noise level in calculation 19.2 is assumed,** not taken from [2], which does not state it. The check is labelled a consistency check for this reason.

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